

THE HODGE CONJECTURE FOR FERMAT VARIETIES OF DEGREE 35

TREVIN PETERSON

ABSTRACT. Let $X_m^n \subset \mathbb{P}^{n+1}$ be the Fermat variety of dimension n and degree m . We prove the Hodge conjecture for X_m^n for every n when $m = 2^a 3^b 5^c 7^d$ is not divisible by 420. For $cd = 0$ this is due to Aoki, and the smallest new degree is $m = 35$. By Aoki's work on the gap group, what is missing is the algebraicity of the Hodge classes attached to one exceptional character of X_{35}^6 . We obtain it from a coniveau statement: the 24-dimensional rational sub-Hodge structure of level one of $H^3(X_{35}^3, \mathbb{Q})$ attached to the character $(1, 2, 16, 21, 30)$ of μ_{35}^5 is supported on a divisor, as predicted by the generalized Hodge conjecture. The proof uses an explicit one-parameter family of curves on X_{35}^3 , a residue formula for the derivative of the Abel–Jacobi map along the family, and a computer-verified interval-arithmetic certificate at one point of the family. For the ten degrees $m = 35k$, $1 \leq k \leq 10$, the reduction to this one class is verified by an exact finite computation which does not use Aoki's structure theorem. As consequences we obtain the Hodge conjecture for products of Fermat varieties of one such degree and for abelian varieties of Fermat type of these degrees, among them the Jacobian of the curve $y^2 = x^{35} - 1$. Finally, for $m = 420$ we show that the Hodge characters in the one remaining class of the gap group have dimension at least 14, a bound which is attained, and deduce the Hodge conjecture for X_{420}^n for $n \leq 12$.

1. INTRODUCTION

Let $X_m^n \subset \mathbb{P}^{n+1}$ be the Fermat variety $x_0^m + \cdots + x_{n+1}^m = 0$. The group $G_m^n = \mu_m^{n+2}/\mu_m$ acts on X_m^n by rescaling the coordinates, and Shioda [24] (see also [22, 25]) described the Hodge classes of X_m^n in terms of its characters: the primitive cohomology $H_{\text{prim}}^n(X_m^n, \mathbb{C})$ is the direct sum of one-dimensional eigenspaces $V(a)$, indexed by an explicit set $\mathfrak{A}_m^n$ of characters a , and for even n the Hodge classes are spanned by the $V(a)$ with a in an explicit subset $\mathfrak{B}_m^n$ (Section 2). The Hodge conjecture for X_m^n is thus the statement that every $V(a)$ with $a \in \mathfrak{B}_m^n$ is spanned by classes of algebraic cycles. It holds for $n = 2$ by the Lefschetz $(1, 1)$ -theorem; for all n when m is prime or $m = 4$ [22, 24], when $m \leq 20$ [24], when m is a prime power [2, Cor. 2-3], when $m = 2p^e$ with p an odd prime or $m = 2^a 3^b 5^c 7^d$ with $cd = 0$ (see the proof of [3, Thm. 0.1]), and when $m \in \{21, 27\}$ [7]; and for $n = 4$ when $m \leq 100$ is prime to 6 [7]; a recent computer-assisted preprint treats $n = 4$ and all odd $m \leq 199$ [16]. For algebraic cycles on self-products of cyclic covers see [23].

The known cases rest on three sources of algebraic cycles: linear subspaces of X_m^n , which account for the decomposable characters $(d_0, -d_0, \dots, d_r, -d_r)$ [22, 24]; the inductive structure of Fermat varieties [25, 24], which relates X_m^n to $X_m^r \times X_m^s$ and to $X_m^{r-1} \times X_m^{s-1}$ for $r + s = n$; and Aoki's cycles [2], which account for his standard characters $\sigma_{p,i}$. Following Aoki [1, 2, 3] we call $a \in \mathfrak{B}_m^n$ *standard* if, up to the order of the entries, $a * d = \sigma_1 * \cdots * \sigma_k * d'$ for decomposable d, d' and standard characters σ_j , where $*$ denotes juxtaposition (Section 2.2); for standard a the space $V(a)$ is spanned by algebraic classes [3, Lemma 4.1]. We call a *exceptional* if it is

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not standard. Exceptional characters exist if and only if m is not of the form p^e or $2p^e$ with p prime [3, Cor. 2.2], and Aoki showed that they are governed by a finite group, the *gap group* B_m/S_m (Section 5): it is a vector space over $\mathbb{F}_2$ with a basis indexed by the divisors $q > 1$ of m which are products of an even number of distinct elements of $\{4\} \cup \{\text{odd primes}\}$ [1, Thm. D], [3, Thm. 2.1, Rem. 2.4], [28]. The Hodge conjecture for X_m^n for all n therefore follows once, for each such q , one Hodge character in the corresponding class is known to be algebraic. For $m = 2^a 3^b 5^c 7^d$ the relevant divisors are among 12, 15, 20, 21, 28, 35 and 420. Aoki gave algebraic Hodge characters, of dimension two or four, in the classes for $q \in \{12, 15, 20, 21\}$, and for $q = 28$ up to a misprint (see the remark after Lemma 5.7); this yields the case $cd = 0$ above. For $q = 35$ no algebraic Hodge character in the class was known. At level 35 the characters in this class occur first in dimension 6 (Remark 5.10), and one of them is

$$\beta = (1, 2, 16, 21, 30, 17, 22, 31) \in \mathfrak{B}_{35}^6.$$

For a character a of G_m^n let $M_a \subset H^n(X_m^n, \mathbb{Q})$ denote the rational Hodge structure spanned by the Galois conjugates of $V(a)$ (Definition 2.2), and let $N^1 H^3(X, \mathbb{Q})$ denote the space of classes supported on a divisor (Section 2.4).

Theorem 1.1. *Let $\alpha = (1, 2, 16, 21, 30)$, a character of G_{35}^3 .*

- (A) $M_\alpha \subset H^3(X_{35}^3, \mathbb{Q})$ is a sub-Hodge structure of dimension 24 with Hodge numbers $h^{2,1} = h^{1,2} = 12$, and $M_\alpha \subset N^1 H^3(X_{35}^3, \mathbb{Q})$. More precisely, there are a smooth projective curve C (not necessarily connected) and an algebraic cycle Z of dimension 2 on $C \times X_{35}^3$ such that $M_\alpha \subset Z_* H^1(C, \mathbb{Q})$.
- (B) For $m \in \{35, 70, 105, 140, 175, 210, 245, 280, 315, 350\}$ and every $n \geq 0$, the Hodge conjecture holds for X_m^n .
- (C) More generally, the Hodge conjecture holds for X_m^n for every $n \geq 0$ and every $m = 2^a 3^b 5^c 7^d$, $a, b, c, d \geq 0$, such that $420 \nmid m$.
- (D) Let m be as in (C). Then the Hodge conjecture holds for every product $X_m^{r_1} \times \cdots \times X_m^{r_k}$, and for every abelian variety of Fermat type of degree m , i.e. every abelian variety isogenous to a factor of a power of the Jacobian of the Fermat curve X_m^1 . In particular, it holds for every power of the Jacobian of the hyperelliptic curve $y^2 = x^{35} - 1$.
- (E) For $m = 420$, the Hodge conjecture holds for X_{420}^n for every $n \leq 12$, and for every product $X_{420}^{r_1} \times \cdots \times X_{420}^{r_k}$ with $r_1 + \cdots + r_k + 2(k-1) \leq 12$.

The step from (A) to (B)–(E) is the algebraicity of the 24-dimensional space $M_\beta \subset H^6(X_{35}^6, \mathbb{Q})$ of Hodge classes, where $\beta = \alpha * (17, 22, 31)$ (Theorem 4.3).

Remark 1.2. (a) Statement (A) is the case of M_α of Grothendieck’s generalized Hodge conjecture [13]: M_α is a sub-Hodge structure of level one of H^3 , and the conjecture predicts that it lies in $N^1 H^3$. If a character a of G_m^3 has two entries with $a_i + a_j \equiv 0$, the analogous statement follows from the inductive structure (Theorem 2.7 with $s = 1$); no two entries of α have this property.

(b) The proof of (B) uses (A), the classical results recalled in Section 2, and an exact finite computation at each of the ten levels $m = 35k$, $1 \leq k \leq 10$ (Proposition 5.8). The proof of (C) uses instead Aoki’s generation theorem for the gap group [3, Thm. 2.1], [1, Thm. D] (Theorem 5.5(i)); the other ingredients are proved here, are classical, or are finite computations at the six levels 12, 15, 20, 21, 28, 35 (Lemma 5.7). Statement (D) follows from (C) by results of Shioda [24, Thm. IV] and Aoki [3, Cor. 3.2]. The proof of (E) (Section 8) uses (A), the same classical results, and elementary identities for character sums at level 420; Aoki’s generation theorem enters only one of two proofs of Proposition 8.3, the other being a finite computation.

(c) If $35 \nmid m$, statement (C) is due to Aoki; the argument of Section 5.5 is his argument in the proof of [3, Thm. 0.1], with the passage from the gap group to characters made explicit

(Lemma 5.3) and with the representatives of Lemma 5.7. Let $35 \mid m$. Then $\mathfrak{B}_m^n$ contains exceptional characters in the class of β in the gap group for every even $n \geq 6$, and also for $n = 4$ if m is even (Remark 5.10); for these pairs (m, n) we are not aware of a previous proof of the Hodge conjecture for X_m^n . For the five odd levels in (B) the class of β contains no character of dimension at most 4, so that the case $n = 4$ of (B) does not depend on (A) there: it follows from Proposition 5.8, Lemma 5.3 and Remark 5.10. For $m \in \{35, 105, 175\}$ this case was proved in [7, 16]; for $m \in \{245, 315\}$ we know no earlier reference. Thus at the ten levels of (B) the cases that depend on (A) are the even $n \geq 6$ for odd m and the even $n \geq 4$ for even m .

(d) The characters $\gamma = (1, 20, 24, 42, 61, 62) \in \mathfrak{B}_{70}^4$ and $\delta = (2, 9, 129, 142, 168, 180) \in \mathfrak{B}_{210}^4$ are among the classes on Fermat fourfolds of even degree at most 250 which are left unresolved in [17, Cor. 6.1] (see also [16, Appendix D]), and it is shown in [17, Thm. 4.1, Prop. 4.2], by an explicit exchange identity, that $V(\delta)$ is algebraic if and only if $V(3\gamma)$ is. Both γ and δ lie in the class of β . In Section 7 we derive the algebraicity of M_γ and M_δ from that of M_β by explicit identities; for δ this is Jumagulov's exchange identity.

(e) Corollary 3.2 of [7] states, as an immediate consequence of [24], the Hodge conjecture for products $X_{m_1}^4 \times \cdots \times X_{m_k}^4$ with $m_i \leq 100$ prime to 6. For $m_1 = m_2 = 35$ this statement implies the algebraicity of M_β : the characters $b_1 = (1, 1, 1, 2, 31, 34)$ and $b_2 = (16, 17, 21, 22, 30, 34)$ lie in $\mathfrak{A}_{35}^4$ and $b_1 * b_2$ equals $\beta * (1, 34, 1, 34)$ up to order, so $V(b_1) \otimes V(b_2)$ consists of Hodge classes on $X_{35}^4 \times X_{35}^4$, and by [24, Thm. II(c2)] and Theorem 2.4 these classes are algebraic if and only if $\beta \in \mathfrak{C}_{35}^6$. The result of [24] on products, Theorem IV there, concerns products of Fermat varieties of one degree m and derives the Hodge conjecture for $X_m^{r_1} \times \cdots \times X_m^{r_k}$ from the Hodge conjecture for X_m^r for all even $r \leq r_1 + \cdots + r_k + 2(k-1)$. For $X_{35}^4 \times X_{35}^4$ this requires, besides $r = 2$ and $r = 4$, the cases $r = 6, 8, 10$, which are not treated in [7] (its results concern X_m^4 , Proposition 3.1, and X_{21}^n, X_{27}^n , Theorem 3.3). These cases are contained in (B), so that Corollary 6.1 gives a proof of the case $m_1 = \cdots = m_k = 35$ of [7, Cor. 3.2]. We do not use [7, Cor. 3.2].

(f) Statement (D) extends [3, Thm. 0.1(i)] from $cd = 0$ to $420 \nmid m$. The Hodge conjecture for the Jacobian of $y^2 = x^{35} - 1$ was left open in [4, Example 8.6], where it is shown that every simple factor of this Jacobian is stably nondegenerate.

(g) If $420 \mid m$, the gap group has one further basis class, and the Hodge conjecture for X_m^n for all n and all $m = 2^a 3^b 5^c 7^d$ reduces to the algebraicity of the Hodge classes attached to a single character of level 420 (Remark 5.11). In Section 8 we show that the Hodge characters of level 420 in this class have dimension at least 14, which gives (E), and that there are such characters of dimension 14 (Theorem 8.1); the reduction can then be made to an explicit character w_1 of dimension 14 (Remark 8.9). The Hodge conjecture for X_{420}^n with even $n \geq 14$, and for X_m^n with even $n \geq 4$ and $m = 420k, k \geq 2$, is not treated here.

Statement (A) is the geometric input, and it is the only step which is not classical. In Section 3 we write down a one-parameter family of curves on X_{35}^3 , defined by an identity among binary forms, prove a residue formula for the derivative of its Abel–Jacobi map in the direction of M_α (Proposition 3.11), and deduce (A) from the nonvanishing of an explicit residue sum at a single point of the family (Theorem 3.16); this nonvanishing is established by a computer-verified certificate in Appendix A. In Section 4 we deduce the algebraicity of M_β from (A) by the Lefschetz $(1, 1)$ -theorem and the inductive structure. Section 5 recalls Aoki's gap group, proves a lemma which passes from generators of the gap group to algebraic cycles in all dimensions, and proves (B) and (C). Section 6 proves (D), Section 7 treats γ and δ , and Section 8 treats the level 420 and proves (E). The identities used are listed in Appendix B; the finite computations are carried out by the ancillary files.

Conventions. Varieties are over $\mathbb{C}$. Cohomology is singular cohomology, and $H^k(Y, \mathbb{C}) = H^k(Y, \mathbb{Q}) \otimes \mathbb{C}$ carries its Hodge decomposition. For a smooth projective variety Y we write $\text{Alg}^p(Y) \subset H^{2p}(Y, \mathbb{Q})$ for the $\mathbb{Q}$ -span of the classes of algebraic cycles of codimension p ; its elements are called algebraic classes.

2. PRELIMINARIES

2.1. Characters and Hodge types. Fix $m \geq 2$ and $n \geq 0$ and put $X = X_m^n$. The group $G_m^n = \mu_m^{n+2}/(\text{diagonal})$ acts on X by $[\zeta_0 : \cdots : \zeta_{n+1}] \cdot (x_0 : \cdots : x_{n+1}) = (\zeta_0 x_0 : \cdots : \zeta_{n+1} x_{n+1})$. Its characters are identified with

$$\widehat{G}_m^n = \{a = (a_0, \dots, a_{n+1}) \in (\mathbb{Z}/m)^{n+2} : a_0 + \cdots + a_{n+1} = 0\}, \quad a(g) = \prod_i \zeta_i^{a_i}.$$

For $a \in \widehat{G}_m^n$ put $V(a) = \{\eta \in H_{\text{prim}}^n(X, \mathbb{C}) : g^* \eta = a(g) \eta \text{ for all } g \in G_m^n\}$, and let $\mathfrak{A}_m^n = \{a \in \widehat{G}_m^n : a_i \neq 0 \text{ for all } i\}$. For $a \in \mathfrak{A}_m^n$ let $|a| = \sum_i \langle a_i \rangle / m$, where $\langle a_i \rangle \in \{1, \dots, m-1\}$ represents a_i , and for $t \in (\mathbb{Z}/m)^\times$ let $ta = (ta_0, \dots, ta_{n+1})$.

Theorem 2.1 (Shioda [24, Thm. I]). (i) For $a \in \widehat{G}_m^n$, $\dim V(a) \leq 1$, with equality if and only if $a \in \mathfrak{A}_m^n$.

(ii) $H_{\text{prim}}^{p,q}(X) = \bigoplus_{|a|=q+1} V(a)$ for $p+q=n$.

(iii) If $n=2p$, then $(H^{p,p} \cap H_{\text{prim}}^n(X, \mathbb{Q})) \otimes \mathbb{C} = \bigoplus_{a \in \mathfrak{B}_m^n} V(a)$, where $\mathfrak{B}_m^n = \{a \in \mathfrak{A}_m^n : |ta| = p+1 \text{ for all } t \in (\mathbb{Z}/m)^\times\}$.

(iv) If $n=2p+1$, then $((H^{p+1,p} \oplus H^{p,p+1}) \cap H^n(X, \mathbb{Q})) \otimes \mathbb{C} = \bigoplus_{a \in \mathfrak{B}_m^n} V(a)$, where $\mathfrak{B}_m^n = \{a \in \mathfrak{A}_m^n : p+1 \leq |ta| \leq p+2 \text{ for all } t \in (\mathbb{Z}/m)^\times\}$.

Definition 2.2. For $a \in \mathfrak{A}_m^n$ let $[a] = \{ta : t \in (\mathbb{Z}/m)^\times\}$ and

$$M_a = H^n(X, \mathbb{Q}) \cap \bigoplus_{b \in [a]} V(b).$$

The idempotent $e_{[a]} = |G_m^n|^{-1} \sum_{b \in [a]} \sum_g \overline{b(g)} g^*$ has coefficients in $\mathbb{Q}(\zeta_m)$ and is invariant under $\text{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q})$, which permutes $[a]$; hence it is defined over $\mathbb{Q}$, and $M_a \otimes \mathbb{C} = \bigoplus_{b \in [a]} V(b)$. In particular $\dim_{\mathbb{Q}} M_a$ is the number of elements of $[a]$, and M_a is a sub-Hodge structure. If n is even and $a \in \mathfrak{B}_m^n$, then M_a consists of Hodge classes, since $[a] \subset \mathfrak{B}_m^n$.

For even n let $\mathfrak{C}_m^n$ be the set of $a \in \mathfrak{B}_m^n$ such that $V(a) \subset \text{Alg}^{n/2}(X) \otimes \mathbb{C}$ [24, p. 177]. The Hodge conjecture for X_m^n is equivalent to $\mathfrak{C}_m^n = \mathfrak{B}_m^n$.

Lemma 2.3. Let n be even and $a \in \mathfrak{B}_m^n$.

- (a) If $t \in (\mathbb{Z}/m)^\times$, then $a \in \mathfrak{C}_m^n$ if and only if $ta \in \mathfrak{C}_m^n$. Moreover $a \in \mathfrak{C}_m^n$ if and only if $M_a \subset \text{Alg}^{n/2}(X)$.
- (b) If a' is obtained from a by permuting the entries, then $a' \in \mathfrak{C}_m^n$ if and only if $a \in \mathfrak{C}_m^n$.
- (c) Let $e \geq 1$, let $f_e : X_{em}^n \rightarrow X_m^n$ be the morphism $(x_i) \mapsto (x_i^e)$, and let $ea \in \mathfrak{A}_{em}^n$ have entries $e \langle a_i \rangle \bmod em$. Then $f_e^* V(a) = V(ea)$; if $a \in \mathfrak{B}_m^n$ then $ea \in \mathfrak{B}_{em}^n$, since $|t(ea)| = |\bar{t}a|$ with $\bar{t} = t \bmod m$; and if $a \in \mathfrak{C}_m^n$ then $ea \in \mathfrak{C}_{em}^n$.

Proof. (a) The maps g^* are defined over $\mathbb{Q}$ and $a(g) \in \mathbb{Q}(\zeta_m)$, so $V(a)$ is spanned by $V(a)_{\mathbb{Q}(\zeta_m)} = V(a) \cap H^n(X, \mathbb{Q}(\zeta_m))$, and the automorphism $\zeta_m \mapsto \zeta_m^t$ of $\mathbb{Q}(\zeta_m)$, acting on coefficients, maps $V(a)_{\mathbb{Q}(\zeta_m)}$ onto $V(ta)_{\mathbb{Q}(\zeta_m)}$ and preserves $\text{Alg}^{n/2}(X) \otimes \mathbb{Q}(\zeta_m)$. This proves the first assertion; the second follows because M_a and $\text{Alg}^{n/2}(X)$ are $\mathbb{Q}$ -subspaces and $M_a \otimes \mathbb{C} = \bigoplus_t V(ta)$. (b) A permutation of the coordinates is an automorphism of X which maps $V(a)$ onto $V(a')$ and preserves algebraic classes. (c) The morphism f_e is equivariant for the homomorphism $G_{em}^n \rightarrow G_m^n$, $[\zeta_i] \mapsto [\zeta_i^e]$. Hence $g^* f_e^* \eta = f_e^* (g^e)^* \eta = a(g^e) f_e^* \eta = (ea)(g) f_e^* \eta$ for $\eta \in V(a)$. Since f_e is

finite and surjective, $f_{e*}f_e^*$ is multiplication by $\deg f_e$, so f_e^* is injective and $f_e^*V(a) = V(ea)$ by Theorem 2.1(i). Finally f_e^* maps algebraic classes to algebraic classes [5], [10, Ch. 19]. $\square$

2.2. Juxtaposition. For $a = (a_0, \dots, a_{r+1}) \in \mathfrak{A}_m^r$ and $b = (b_0, \dots, b_{s+1}) \in \mathfrak{A}_m^s$ let $a * b = (a_0, \dots, a_{r+1}, b_0, \dots, b_{s+1}) \in \mathfrak{A}_m^{r+s+2}$. Clearly $|t(a * b)| = |ta| + |tb|$, so $a * b \in \mathfrak{B}_m^{r+s+2}$ if $a \in \mathfrak{B}_m^r$ and $b \in \mathfrak{B}_m^s$ with r, s even. For even r let $\mathfrak{D}_m^r \subset \mathfrak{B}_m^r$ be the set of characters which, up to the order of the entries, have the form $(d_0, -d_0, d_1, -d_1, \dots, d_{r/2}, -d_{r/2})$ with $d_i \in \mathbb{Z}/m \setminus \{0\}$. For a prime p dividing m , $d = m/p$ and an integer i with $0 < i < d$, Aoki's *standard characters* are

$$\sigma_{p,i} = (i, i + d, i + 2d, \dots, i + (p-1)d, -pi) \quad (p \geq 3), \quad \sigma_{2,i} = (i, i + d, -2i, d)$$

[3, Section 2] (see also [2, p. 387]); a character $a \in \mathfrak{B}_m^n$ is *standard* if $a * d \sim \sigma_1 * \dots * \sigma_k * d'$ for some $d, d' \in \bigcup_r \mathfrak{D}_m^r$ ($r \geq 0$ even) and standard characters σ_j , where $\sim$ means equality up to the order of the entries. This is the definition of [3, Section 2], except that there d, d' are taken in $\mathfrak{D}_m^r$ with $r > 0$. In [2, p. 387] the standard characters are restricted to $d/(i, d) > 2$, and decomposable factors enter only in [2, Thms. 1-2, 1-3]. All these variants give the same sets of standard characters: the excluded characters $\sigma_{p,d/2}$ (d even) are decomposable, and a decomposable factor in $\mathfrak{D}_m^0$ can be enlarged by a pair $(e, -e)$ on both sides.

Theorem 2.4 (Shioda, Ran; see [2, Thm. 1-4]). *Let $r, s \geq 0$ be even, $a \in \mathfrak{B}_m^r$ and $b \in \mathfrak{B}_m^s$.*

- (i) *If $a \in \mathfrak{C}_m^r$ and $b \in \mathfrak{C}_m^s$, then $a * b \in \mathfrak{C}_m^{r+s+2}$.*
- (ii) *If $d \in \mathfrak{D}_m^s$ and $a * d \in \mathfrak{C}_m^{r+s+2}$, then $a \in \mathfrak{C}_m^r$.*

Part (i) is [24, Corollary to Thm. II, (i)]; part (ii) is the case $b \in \mathfrak{D}_m^s$ of [24, Lemma 3] together with $\mathfrak{D}_m^s \subset \mathfrak{C}_m^s$ [2, Thm. 1-1]; see also [3, Lemma 4.1].

Theorem 2.5 (Aoki [2, Thm. 2-1]). *Let $p \geq 3$ be a prime, $m = pd$, and let i be an integer with $0 < i < d$ and $\gcd(i, d) = 1$. Then $\sigma_{p,i} \in \mathfrak{C}_m^{p-1}$.*

Theorem 2.6 (Lefschetz; [24, (1.10)], [11, p. 163]). $\mathfrak{C}_m^2 = \mathfrak{B}_m^2$.

2.3. The inductive structure.

Theorem 2.7 (Shioda–Katsura [24, (2.1), Thm. II, (2.14)], [25]). *Let $r, s \geq 1$, $n = r + s$ and $Y = X_m^{r-1} \times X_m^{s-1}$. Let $\text{bl}: W \rightarrow X_m^r \times X_m^s$ be the blowing up along Y , embedded by $((x_i), (y_j)) \mapsto ((x_0 : \dots : x_r : 0), (y_0 : \dots : y_s : 0))$, let $j: \tilde{Y} = \text{bl}^{-1}(Y) \rightarrow W$ be the inclusion and $\text{bl}_Y: \tilde{Y} \rightarrow Y$ the restriction of bl . There is a generically finite surjective morphism $\psi: W \rightarrow X_m^n$ such that the homomorphism*

$$f = \psi_* \circ j_* \circ \text{bl}_Y^*: H^{n-2}(Y, \mathbb{Q}) \rightarrow H^n(X_m^n, \mathbb{Q})$$

*maps $V(a') \otimes V(b')$ onto $V(a' * b')$ for all $a' \in \mathfrak{A}_m^{r-1}$ and $b' \in \mathfrak{A}_m^{s-1}$. In particular, if n is even and $V(a') \otimes V(b') \subset \text{Alg}^{(n-2)/2}(Y) \otimes \mathbb{C}$, then $a' * b' \in \mathfrak{C}_m^n$.*

Here ψ is the composition of bl with Shioda's rational map $((x_i), (y_j)) \mapsto (x_0 y_{s+1} : \dots : x_r y_{s+1} : \epsilon x_{r+1} y_0 : \dots : \epsilon x_{r+1} y_s)$, $\epsilon^m = -1$, and f is the map which Shioda obtains on the second summand of [24, (2.4)] (see the proof of Theorem II there). The last assertion holds because pullbacks and pushforwards along morphisms of smooth projective varieties preserve algebraic classes [5], [10, Ch. 19].

2.4. Correspondences and coniveau. Let C be a smooth projective curve (not necessarily connected), X' a smooth projective threefold and Z an algebraic cycle of dimension 2 on $C \times X'$ with rational coefficients. Put $Z_*(b) = \text{pr}_{X'*}([Z] \cup \text{pr}_C^* b)$ and $Z^*(x) = \text{pr}_{C*}([Z] \cup \text{pr}_{X'}^* x)$. Then $Z_*: H^1(C, \mathbb{Q}) \rightarrow H^3(X', \mathbb{Q})$ and $Z^*: H^3(X', \mathbb{Q}) \rightarrow H^1(C, \mathbb{Q})$ are morphisms of Hodge

structures of bidegree $(1, 1)$ and $(-1, -1)$, i.e. morphisms $H^1(C, \mathbb{Q})(-1) \rightarrow H^3(X', \mathbb{Q})$ and $H^3(X', \mathbb{Q}) \rightarrow H^1(C, \mathbb{Q})(-1)$, and

$$(2.1) \quad \int_{X'} Z_*(b) \cup x = \int_{C \times X'} [Z] \cup \text{pr}_C^* b \cup \text{pr}_{X'}^* x = \int_C b \cup Z^*(x) \quad (b \in H^1(C), x \in H^3(X')).$$

If $Z = (p, q)_*[S]$ for a smooth projective surface S and morphisms $p: S \rightarrow C$, $q: S \rightarrow X'$, then $Z_* = q_* p^*$ and $Z^* = p_* q^*$. Compositions of correspondences are induced by algebraic cycles and preserve algebraic classes [5], [10, Ch. 16, 19].

Let $N^1 H^3(X', \mathbb{Q}) = \sum_D \ker(H^3(X', \mathbb{Q}) \rightarrow H^3(X' \setminus D, \mathbb{Q}))$, the sum over closed algebraic subsets $D \subset X'$ of codimension at least one.

Lemma 2.8. *In this situation $Z_* H^1(C, \mathbb{Q}) \subset N^1 H^3(X', \mathbb{Q})$.*

Proof. Let D be the image in X' of the support of Z ; it has dimension at most 2. The class $[Z] \cup \text{pr}_C^* b$ comes from cohomology of $C \times X'$ with supports in $C \times D$, so $Z_*(b)$ comes from cohomology of X' with supports in D and restricts to zero on $X' \setminus D$. $\square$

3. A FAMILY OF CURVES ON X_{35}^3 AND A PIECE OF CONIVEAU ONE

3.1. Notation. Throughout this section $X = X_{35}^3 \subset \mathbb{P}^4$ with equation $F = x_0^{35} + \dots + x_4^{35}$, $F_i = \partial F / \partial x_i = 35x_i^{34}$, $G = G_{35}^3$ (a group of order 35^4), $U_i = X \cap \{x_i \neq 0\}$ (affine and G -stable), and $\alpha = (1, 2, 16, 21, 30)$. Let

$$\mathcal{T} = \{1, 2, 3, 4, 6, 9, 11, 12, 18, 19, 22, 27\} \subset (\mathbb{Z}/35)^\times.$$

Lemma 3.1. (a) $\alpha \in \mathfrak{A}_{35}^3$, and $|t\alpha| = 2$ for $t \in \mathcal{T}$, $|t\alpha| = 3$ for $t \notin \mathcal{T}$. Hence $\alpha \in \mathfrak{B}_{35}^3$, $V(t\alpha) \subset H^{2,1}(X)$ for $t \in \mathcal{T}$ and $V(t\alpha) \subset H^{1,2}(X)$ otherwise, and M_α has dimension 24 with $h^{2,1} = h^{1,2} = 12$. (b) $[\alpha]$ consists of 24 distinct characters, and $-\alpha = 34\alpha \in [\alpha]$. (c) No two entries of α have sum 0 in $\mathbb{Z}/35$.

Proof. Direct computation (Table 2) and Theorem 2.1. $\square$

In particular $V(\alpha)$ is the α -eigenspace of $H^{2,1}(X)$, and it is one-dimensional. Let

$$\Pi = \{z \in \mathbb{P}^4 : z_0 + \dots + z_4 = 0\}, \quad \Phi: X \rightarrow \Pi, \quad [x] \mapsto [x_0^{35} : \dots : x_4^{35}].$$

The morphism Φ is finite and surjective, identifies Π with X/G , and is étale over $\Pi^\circ = \Pi \cap \{z_0 z_1 z_2 z_3 z_4 \neq 0\}$.

3.2. The family. Let $y = (p_0, \dots, p_4, q_0, q_1, b, c_1, \dots, c_4)$ be coordinates on $\mathbb{A}^{12}$ and consider the binary forms

$$P = s^5 + p_4 s^4 t + p_3 s^3 t^2 + p_2 s^2 t^3 + p_1 s t^4 + p_0 t^5, \quad Q = s^2 + q_1 s t + q_0 t^2, \quad R = s^2 + b s t + t^2,$$

and the five binary forms of degree 10

$$(3.1) \quad u_0 = t^3 P Q, \quad u_1 = c_1 t P Q^2, \quad u_2 = c_2 P^2, \quad u_3 = c_3 R^5, \quad u_4 = c_4 s^7 t Q.$$

Let $\mathcal{P} \subset \mathbb{A}^{12}$ be the closed subscheme on which $u_0 + u_1 + u_2 + u_3 + u_4 = 0$ identically in (s, t) . Writing $P(s) = P(s, 1)$ and similarly for the other forms, $\mathcal{P}$ is defined by the vanishing of the eleven coefficients $E_0, \dots, E_{10}$ of

$$(3.2) \quad E_y(s) = P(s)Q(s) + c_1 P(s)Q(s)^2 + c_2 P(s)^2 + c_3 R(s)^5 + c_4 s^7 Q(s),$$

a polynomial of degree at most 10 in s . We write $J_b(y)$ for the 11×11 Jacobian of $(E_0, \dots, E_{10})$ with respect to the eleven coordinates other than b , and $T_y \mathcal{P} \subset \mathbb{C}^{12}$ for the Zariski tangent space.

Definition 3.2. A point $y \in \mathcal{P}$ is *nondegenerate* if $c_1 c_2 c_3 c_4 \neq 0$, $\text{disc}(R) = b^2 - 4 \neq 0$, $\text{Res}(R, P) \neq 0$ and $\text{Res}(R, Q) \neq 0$.

Lemma 3.3. Let $y \in \mathcal{P}$ be nondegenerate. (a) The forms u_3 and u_4 have no common zero on $\mathbb{P}^1$; hence $f_y = [u_0 : \cdots : u_4] : \mathbb{P}^1 \rightarrow \Pi$ is a morphism. (b) The zeros of u_3 are the two distinct roots $\rho_1, \rho_2 \in \mathbb{C}$ of $R(s)$, and s , $P(s)$ and $Q(s)$ do not vanish at ρ_1, ρ_2 .

Proof. The zeros of u_4 are $s = 0$, $t = 0$ and the zeros of Q , while u_3 vanishes exactly at the zeros of R ; now $R(0, 1) = R(1, 0) = 1$ and $\text{Res}(R, Q) \neq 0$. Part (b) follows from $\text{disc}(R) \neq 0$, $R(0) = 1$ and $\text{Res}(R, P) \text{Res}(R, Q) \neq 0$. $\square$

For nondegenerate y let

$$\mathcal{C}_y = \mathbb{P}^1 \times_{\Pi} X = \{((s : t), [x]) \in \mathbb{P}^1 \times X : [x_0^{35} : \cdots : x_4^{35}] = [u_0(s, t) : \cdots : u_4(s, t)]\},$$

a curve which is finite over $\mathbb{P}^1$. For an irreducible component of $\mathcal{C}_y$ we denote by $\widehat{C}$ its normalization and by $\pi_1 : \widehat{C} \rightarrow \mathbb{P}^1$ and $\nu : \widehat{C} \rightarrow X$ the two projections; π_1 is finite of some degree $\deg \pi_1 \geq 1$. Let $\widehat{C}^\circ = \pi_1^{-1}\{(s : 1) : u_0 \cdots u_4(s, 1) \neq 0\}$. Since Φ is étale over Π° , near every point of $\widehat{C}^\circ$ the function s is a local coordinate and ν is given by $s \mapsto [x(s, y)]$ with

$$(3.3) \quad x(s, y') = (u_0(s, 1; y')^{1/35}, \dots, u_4(s, 1; y')^{1/35})$$

for suitable holomorphic branches, defined for $y' \in \mathbb{C}^{12}$ near y .

In the product $u_0 u_1^2 u_2^{16} u_3^{21} u_4^{30}$ the forms s, t, R, P, Q occur with exponents 210, 35, 105, 35, 35. Hence, with

$$(3.4) \quad A = s^6 t R^3 P Q, \quad \Psi(y) = c_1^2 c_2^{16} c_3^{21} c_4^{30},$$

we have the identity of binary forms

$$(3.5) \quad u_0 u_1^2 u_2^{16} u_3^{21} u_4^{30} = \Psi(y) A^{35}.$$

3.3. The residue functional.

Definition 3.4. Let $y \in \mathcal{P}$ be nondegenerate and $v \in \mathbb{C}^{12}$. Denote by a dot the derivative along v at fixed s , so that $\dot{P}(s) = \sum_k v_{p_k} s^k$, $\dot{Q}(s) = v_{q_0} + v_{q_1} s$ and $\dot{c}_k = v_{c_k}$, and by a prime the derivative in s . Put

$$\Delta_v(s) = \left(\frac{\dot{c}_1}{c_1} + \frac{\dot{Q}}{Q} \right) \left(\frac{P'}{P} - \frac{Q'}{Q} \right) - \left(\frac{\dot{c}_2}{c_2} + \frac{\dot{P}}{P} - \frac{\dot{Q}}{Q} \right) \frac{Q'}{Q}, \quad \eta_v = \frac{P(s) \Delta_v(s)}{c_3 c_4 s R(s)^2} ds,$$

and

$$\mathcal{I}(y, v) = \text{Res}_{s=\rho_1} \eta_v + \text{Res}_{s=\rho_2} \eta_v,$$

where ρ_1, ρ_2 are the roots of $R(s)$.

A direct computation shows that, in the chart $t = 1$,

$$(3.6) \quad \Delta_v = \det \begin{pmatrix} 1 & 1 & 1 \\ \dot{u}_0/u_0 & \dot{u}_1/u_1 & \dot{u}_2/u_2 \\ u'_0/u_0 & u'_1/u_1 & u'_2/u_2 \end{pmatrix}, \quad \eta_v = \frac{A \Delta_v}{u_3 u_4} ds.$$

By Lemma 3.3(b) the poles of η_v at ρ_1, ρ_2 have order at most two and come only from the factor R^{-2} . The quantity $\mathcal{I}(y, v)$ is linear in v , and it depends holomorphically on (y, v) on the set of nondegenerate y (locally, ρ_1, ρ_2 are holomorphic functions of y).

3.4. The Carlson–Griffiths cocycle. Let $\xi = \sum_k x_k \partial / \partial x_k$ be the Euler vector field on $\mathbb{C}^5$, $dV = dx_0 \wedge \cdots \wedge dx_4$ and $\Omega = \iota_\xi dV = \sum_k (-1)^k x_k dx_0 \wedge \cdots \wedge \widehat{dx_k} \wedge \cdots \wedge dx_4$. For $i \neq j$ put $\Omega_{ij} = \iota_{\partial_j} \iota_{\partial_i} \Omega$, where $\partial_i = \partial / \partial x_i$, and

$$\varphi_{ij} = \frac{x^{\alpha-1} \Omega_{ij}}{F_i F_j}, \quad x^{\alpha-1} = \prod_{k=0}^4 x_k^{\alpha_k-1},$$

with $\alpha_k \in \{1, \dots, 34\}$ the entries of α . Note $\Omega_{ji} = -\Omega_{ij}$ and $\varphi_{ji} = -\varphi_{ij}$.

Lemma 3.5. (a) φ_{ij} is the pullback of a rational 2-form on $\mathbb{P}^4$ which is regular on $\{x_i x_j \neq 0\}$; its restriction to X is a regular 2-form on $U_i \cap U_j$. (b) $\varphi_{jk} - \varphi_{ik} + \varphi_{ij} = 0$ on $U_i \cap U_j \cap U_k$. (c) $g^* \varphi_{ij} = \alpha(g) \varphi_{ij}$ for $g \in G$. Consequently $\varphi = \{\varphi_{ij}\}$ is a Čech 1-cocycle of Ω_X^2 for the cover $\{U_i\}$, and its class $[\varphi] \in H^1(X, \Omega_X^2)$ lies in the α -eigenspace.

Proof. (a) Under $x \mapsto \lambda x$ the forms $x^{\alpha-1}$, Ω_{ij} and $F_i F_j$ are multiplied by λ^{65} , λ^3 and λ^{68} , since $\sum_k (\alpha_k - 1) = 65$; so φ_{ij} is invariant. Moreover $\iota_\xi \Omega_{ij} = \iota_\xi \iota_{\partial_j} \iota_{\partial_i} \iota_\xi dV = 0$, since contractions anticommute and $\iota_\xi \iota_\xi = 0$. Hence φ_{ij} is the pullback of a 2-form on $\mathbb{P}^4$, regular where $x_i x_j \neq 0$.

(b) Since $\xi F = 35F$, we have $dF \wedge \Omega = (\iota_\xi dF) dV - \iota_\xi (dF \wedge dV) = 35F dV$. Contracting with $\partial_i, \partial_j, \partial_k$ and using that ι_∂ is an antiderivation with $\iota_{\partial_l} dF = F_l$, one obtains

$$F_i \Omega_{jk} - F_j \Omega_{ik} + F_k \Omega_{ij} = dF \wedge \Omega_{ijk} + 35F \iota_{\partial_k} \iota_{\partial_j} \iota_{\partial_i} dV, \quad \Omega_{ijk} = \iota_{\partial_k} \Omega_{ij}.$$

On the cone over X both F and dF restrict to zero. Multiplying by $x^{\alpha-1}/(F_i F_j F_k)$ gives (b) on the cone, hence on X .

(c) For $g = [\zeta_0 : \cdots : \zeta_4]$ we have $g^* x^{\alpha-1} = \prod \zeta_k^{\alpha_k-1} x^{\alpha-1}$, $g^* F_i = \zeta_i^{-1} F_i$, and every term of Ω_{ij} is a multiple of $x_k dx_l \wedge dx_{l'}$ with $\{k, l, l'\} = \{0, \dots, 4\} \setminus \{i, j\}$, so $g^* \Omega_{ij} = \prod_{k \neq i, j} \zeta_k \Omega_{ij}$. The product of the factors is $\prod_k \zeta_k^{\alpha_k} = \alpha(g)$. $\square$

The cocycle φ is the one attached by Carlson and Griffiths [6, Section 3(b)] to the residue of $x^{\alpha-1} \Omega / F^2$; we shall not need their normalization.

3.5. First-order deformations and a trace formula.

Definition 3.6. Let $y \in \mathcal{P}$ be nondegenerate, $v \in T_y \mathcal{P}$, and let $\widehat{C}, \pi_1, \nu$ be as in Section 3.2. Since $v \in T_y \mathcal{P}$, the vector $\partial x(s, y) / \partial v \in \mathbb{C}^5$ (derivative of (3.3) along v at fixed s) is tangent to the cone over X , because $\partial_v (\sum_k u_k(s, 1)) = \partial_v E_y(s) = 0$; hence it defines a tangent vector to X at the point $[x(s, y)]$. A *first-order deformation of ν along v* is a family of holomorphic sections w_j of $\nu^* T_X$ over the members O_j of an open cover of $\widehat{C}$ such that $w_j - w_{j'} \in d\nu(T_{\widehat{C}})$ on $O_j \cap O_{j'}$ and $w_j \equiv \partial x / \partial v \pmod{d\nu(T_{\widehat{C}})}$ on $O_j \cap \widehat{C}^\circ$.

For such w and a smooth form h of type $(2, 0)$ or $(2, 1)$ on an open subset of X , the form $\nu^*(\iota_w h)$, defined locally as $\nu^*(\iota_{w_j} h)$, is independent of the choices: if τ is tangent to $\widehat{C}$, then $\nu^*(\iota_{d\nu(\tau)} h) = \iota_\tau \nu^* h = 0$, because $\nu^* h$ is a form of type $(2, 0)$, or a form of degree 3, on a curve. Since the w_j are holomorphic, $\bar{\partial} \nu^*(\iota_w h) = -\nu^*(\iota_w \bar{\partial} h)$ for h of type $(2, 0)$.

Lemma 3.7. Let $\widehat{C}$ be a compact Riemann surface covered by open sets W_3, W_4 with $\widehat{C} \setminus W_3$ finite. Let Θ be a smooth 2-form on $\widehat{C}$ and g_3, g_4 smooth $(1, 0)$ -forms on W_3, W_4 with $\bar{\partial} g_i = \Theta$ on W_i , such that $g_4 - g_3$ is holomorphic on $W_3 \cap W_4$. Then

$$\int_{\widehat{C}} \Theta = 2\pi i \sum_{o \in \widehat{C} \setminus W_3} \text{Res}_o(g_4 - g_3).$$

Proof. Choose small disjoint discs $D_o \subset W_4$ centred at the points $o \in \widehat{C} \setminus W_3$. On a curve $\partial g_i = 0$, so $\Theta = dg_i$ on W_i . By Stokes' theorem $\int_{\widehat{C} \setminus \bigcup D_o} \Theta = -\sum_o \int_{\partial D_o} g_3$ and $\int_{D_o} \Theta = \int_{\partial D_o} g_4$. Adding gives $\sum_o \int_{\partial D_o} (g_4 - g_3) = 2\pi i \sum_o \text{Res}_o(g_4 - g_3)$. $\square$

We recall the Čech description of Dolbeault cohomology. If θ is a $\bar{\partial}$ -closed $(2, 1)$ -form on X , there are smooth $(2, 0)$ -forms χ_i on the Stein open sets U_i with $\bar{\partial}\chi_i = \theta|_{U_i}$; then $h_{ij} = \chi_j - \chi_i$ is a Čech 1-cocycle of Ω_X^2 , and $\theta \mapsto \{h_{ij}\}$ induces the Dolbeault isomorphism $H_{\bar{\partial}}^{2,1}(X) \cong H^1(X, \Omega_X^2)$, which is compatible with the action of G [11, Chapter 0, Section 3], [26, Ch. 4]. (The cover $\{U_i\}$ computes $H^1(X, \Omega_X^2)$ because the U_i are affine.)

Lemma 3.8. *Let w be a first-order deformation as in Definition 3.6 and put $W_i = \nu^{-1}(U_i)$. For a Čech 1-cocycle h of Ω_X^2 on $\{U_i\}$ let*

$$\mathcal{R}_w(h) = \sum_{o \in \widehat{C} \setminus W_3} \text{Res}_o \nu^*(\iota_w h_{34}).$$

Then $\mathcal{R}_w(h)$ depends only on the class of h in $H^1(X, \Omega_X^2)$, and for every $\bar{\partial}$ -closed $(2, 1)$ -form θ with associated cocycle h ,

$$\int_{\widehat{C}} \nu^*(\iota_w \theta) = -2\pi i \mathcal{R}_w(h).$$

Proof. By Lemma 3.3(a), x_3 and x_4 have no common zero on $\nu(\widehat{C})$, so $W_3 \cup W_4 = \widehat{C}$, and $\widehat{C} \setminus W_3$ is finite. Put $g_i = \nu^*(\iota_w \chi_i)$ on W_i ($i = 3, 4$). Then $\bar{\partial}g_i = -\nu^*(\iota_w \theta)$ and $g_4 - g_3 = \nu^*(\iota_w h_{34})$ is holomorphic on $W_3 \cap W_4$, so Lemma 3.7 with $\Theta = -\nu^*(\iota_w \theta)$ gives the formula. If h is a coboundary, $h_{ij} = \chi_j - \chi_i$ with χ_i holomorphic, and the same argument with $\theta = 0$ gives $\mathcal{R}_w(h) = 0$. $\square$

Lemma 3.9. *On $\widehat{C}^\circ$, in the local coordinate s ,*

$$\nu^*(\iota_w \varphi_{34}) = 35^{-4} \cdot \nu^*\left(\frac{x^\alpha}{x_3^{35} x_4^{35}}\right) \cdot \Delta_v(s) ds, \quad x^\alpha = \prod_k x_k^{\alpha_k}.$$

Proof. Work on the cone with the lift (3.3). The tangent vector and the deformation vector are

$$\frac{\partial x}{\partial s} = \left(\frac{x_k u'_k}{35 u_k}\right)_k, \quad \frac{\partial x}{\partial v} = \left(\frac{x_k \dot{u}_k}{35 u_k}\right)_k.$$

Since φ_{34} is basic, its value on the images of these vectors may be computed on the cone, and

$$\begin{aligned} \Omega_{34}\left(\frac{\partial x}{\partial v}, \frac{\partial x}{\partial s}\right) &= dV\left(\xi, \partial_3, \partial_4, \frac{\partial x}{\partial v}, \frac{\partial x}{\partial s}\right) \\ &= \det \begin{pmatrix} x_0 & x_1 & x_2 \\ x_0 \dot{u}_0/35 u_0 & x_1 \dot{u}_1/35 u_1 & x_2 \dot{u}_2/35 u_2 \\ x_0 u'_0/35 u_0 & x_1 u'_1/35 u_1 & x_2 u'_2/35 u_2 \end{pmatrix} = \frac{x_0 x_1 x_2}{35^2} \Delta_v, \end{aligned}$$

by expansion along the rows of ∂_3, ∂_4 and (3.6). Dividing by $F_3 F_4 = 35^2 x_3^{34} x_4^{34}$ and multiplying by $x^{\alpha-1}$ gives the claim, since $w \equiv \partial x / \partial v$ modulo tangent vectors of $\widehat{C}$. $\square$

Lemma 3.10. *There is $r \in \mathbb{C}$ with $r^{35} = \Psi(y)$ such that $\nu^*(x^\alpha / (x_3^{35} x_4^{35})) = r \cdot \pi_1^*(A / (u_3 u_4))$ on $\widehat{C}$.*

Proof. Both sides are rational functions on $\widehat{C}$ ($x^\alpha / (x_3^{35} x_4^{35})$ has degree 0 on $\mathbb{P}^4$ and $A / (u_3 u_4)$ has degree 0 in (s, t)). On $\widehat{C}^\circ$, using (3.3) and (3.5), the 35th power of their quotient is $\prod_k u_k^{\alpha_k} / A^{35} = \Psi(y)$. Hence the quotient is constant on the connected curve $\widehat{C}$, and $\Psi(y) \neq 0$. $\square$

Proposition 3.11 (Residue formula). *Let ω be the harmonic representative of a nonzero class $\omega_\alpha \in V(\alpha)$; it is a closed form of type $(2, 1)$. Let $\lambda \in \mathbb{C}$ be defined by $[\varphi] = \lambda[\omega]$ in $H^1(X, \Omega_X^2)$. Then, for every nondegenerate $y \in \mathcal{P}$, $v \in T_y \mathcal{P}$, normalized component $\widehat{C}$ of $\mathcal{C}_y$ and first-order deformation w of ν along v ,*

$$\lambda \int_{\widehat{C}} \nu^*(\iota_w \omega) = -2\pi i \cdot 35^{-4} \cdot r \cdot \deg(\pi_1) \cdot \mathcal{I}(y, v),$$

with r as in Lemma 3.10. In particular, if $\mathcal{I}(y, v) \neq 0$ for one such datum, then $\lambda \neq 0$, and for every datum $\int_{\widehat{C}} \nu^(\iota_w \omega) \neq 0$ if and only if $\mathcal{I}(y, v) \neq 0$.*

Proof. The Dolbeault class of ω is nonzero, and by Theorem 2.1 and Lemma 3.1 the α -eigenspace of $H^1(X, \Omega_X^2) \cong H^{2,1}(X)$ is one-dimensional; by Lemma 3.5(c) it contains $[\varphi]$. So λ is well defined. Let h be the cocycle attached to ω . By Lemma 3.8, $\int_{\widehat{C}} \nu^*(\iota_w \omega) = -2\pi i \mathcal{R}_w(h)$ and $\mathcal{R}_w(\varphi) = \lambda \mathcal{R}_w(h)$. By Lemmas 3.9 and 3.10 and (3.6), the holomorphic 1-form $\nu^*(\iota_w \varphi_{34})$ on $W_3 \cap W_4$ and the meromorphic 1-form $35^{-4} r \pi_1^* \eta_v$ agree on $\widehat{C}^\circ$, hence on the connected set $W_3 \cap W_4$. (Near the points of $\widehat{C}$ over ρ_1, ρ_2 , which lie outside $\widehat{C}^\circ$, the vector $\partial x / \partial v$ need not be holomorphic, and w differs from it by a tangent vector of $\widehat{C}$; this is why the identity is proved on $\widehat{C}^\circ$ and then extended to $W_3 \cap W_4$, which contains a punctured neighbourhood of each of these points.) The points of $\widehat{C} \setminus W_3$ are the points where $x_3 \circ \nu$ vanishes, i.e. $\pi_1^{-1}\{\rho_1, \rho_2\}$ (Lemma 3.3(b)). For a meromorphic 1-form η near a point $\rho \in \mathbb{P}^1$ one has $\sum_{o \in \pi_1^{-1}(\rho)} \text{Res}_o \pi_1^* \eta = \deg(\pi_1) \text{Res}_\rho \eta$, since π_1 is finite. Therefore

$$\mathcal{R}_w(\varphi) = 35^{-4} r \deg(\pi_1) (\text{Res}_{\rho_1} \eta_v + \text{Res}_{\rho_2} \eta_v) = 35^{-4} r \deg(\pi_1) \mathcal{I}(y, v).$$

Combining the three identities gives the formula. The last assertion follows since $r \neq 0$ and $\deg \pi_1 \geq 1$. $\square$

Remark 3.12. The formula concerns a single component. If $g \in G$, then $g \circ \nu$ is the normalization of a component of $\mathcal{C}_y$ (with the same π_1), $dg(w)$ is a first-order deformation of it, and r is replaced by $\alpha(g)r$, in accordance with $g^* \omega = \alpha(g)\omega$. Since the image of $\mathcal{C}_y$ in X is G -stable and α is a nontrivial character, the contributions of all components of $\mathcal{C}_y$ add up to zero; for this reason the argument below works with one component at a time.

Remark 3.13. Up to sign, $\int_{\widehat{C}} \nu^*(\iota_w \omega)$ is the derivative of the Abel–Jacobi map of the family of curves, paired with ω [12, (6.41)]; we shall only use the fibre-integration form of this statement (proof of Proposition 3.15). The constant λ can be computed from [6, Section 3(b)]; the argument above needs only $\lambda \neq 0$, which it proves directly from the nonvanishing of one value of $\mathcal{I}$.

3.6. The family over a component of $\mathcal{P}$. Let $y^* \in \mathcal{P}$ be nondegenerate with $J_b(y^*)$ invertible. By the Jacobian criterion, $\mathcal{P}$ is smooth of dimension one at y^* ; let $\mathcal{P}_1$ be the unique irreducible component of $\mathcal{P}$ through y^* , a curve on which b is a local coordinate at y^* . Let $\mathcal{P}_1^\circ \subset \mathcal{P}_1$ be the Zariski-open set of smooth nondegenerate points y with $J_b(y)$ invertible, and for $y \in \mathcal{P}_1^\circ$ let $v(y) \in T_y \mathcal{P}_1$ be the tangent vector with $v(y)_b = 1$; it depends holomorphically on y .

Lemma 3.14. *There are a smooth connected projective curve B with a nonconstant morphism $\pi: B \rightarrow \overline{\mathcal{P}}_1$ (closure in $\mathbb{P}^{12}$), a smooth projective surface S with morphisms $p: S \rightarrow B$ and $q: S \rightarrow X$, and a nonempty Zariski-open set $B^\circ \subset B$ such that for every $b' \in B^\circ$:*

- (i) $\pi(b') \in \mathcal{P}_1^\circ$ and π is étale at b' , and p is smooth over B° ;
- (ii) the fibre $S_{b'} = p^{-1}(b')$, together with the restrictions of q and of the projection $S \rightarrow \mathbb{P}^1$, is the normalization of an irreducible component of $\mathcal{C}_{\pi(b')}$ with its maps ν and π_1 ;
- (iii) for $v' \in T_{b'} B$, the images under dq of local holomorphic lifts of v' to S form a first-order deformation of ν along $d\pi(v')$.

Proof. Let $\mathcal{C} = \{(y, (s : t), [x]) \in \mathcal{P}_1^\circ \times \mathbb{P}^1 \times X : f_y(s : t) = \Phi([x])\}$. The morphism Φ is finite and flat (a finite morphism of smooth varieties of the same dimension), so its base change $\mathcal{C} \rightarrow \mathcal{P}_1^\circ \times \mathbb{P}^1$ is finite and flat; hence every irreducible component of $\mathcal{C}$ is a surface which maps onto $\mathcal{P}_1^\circ \times \mathbb{P}^1$. Fix one such component $\mathcal{C}'$. Let B be the smooth projective curve whose function field is the algebraic closure of $\mathbb{C}(\mathcal{P}_1)$ in $\mathbb{C}(\mathcal{C}')$, with the induced morphism π . The inclusion of function fields defines a rational map from $\mathcal{C}'$ to B ; together with the projections to $\mathbb{P}^1$ and X it maps $\mathcal{C}'$ birationally onto a dense open subset of an irreducible closed surface $\mathcal{C}'' \subset B \times \mathbb{P}^1 \times X$. As $\mathbb{C}(B)$ is algebraically closed in $\mathbb{C}(\mathcal{C}'')$, the generic fibre of $\mathcal{C}'' \rightarrow B$ is geometrically irreducible. Let S be a projective resolution of singularities of $\mathcal{C}''$ which is an isomorphism over the smooth locus of $\mathcal{C}''$, with the induced morphisms p, q and $S \rightarrow \mathbb{P}^1$. Then $\mathbb{C}(S) = \mathbb{C}(\mathcal{C}'')$, so the generic fibre of $p: S \rightarrow B$ is geometrically irreducible as well, and the exceptional locus of $S \rightarrow \mathcal{C}''$ has only finitely many components which are vertical for p . By [8, 9.7.7] and generic smoothness [14, III.10.7] there is a nonempty open $B^\circ \subset B$ satisfying (i) such that the fibres $S_{b'}$, $b' \in B^\circ$, are smooth, connected and birational to the fibres $\mathcal{C}_{b'}''$, and each $\mathcal{C}_{b'}''$ is an irreducible curve. Since $\mathcal{C}'$ is closed in $\mathcal{P}_1^\circ \times \mathbb{P}^1 \times X$, the image of $\mathcal{C}_{b'}''$ in $\mathbb{P}^1 \times X$ is contained in $\mathcal{C}_{\pi(b')}$; it is finite over $\mathbb{P}^1$, hence an irreducible component of $\mathcal{C}_{\pi(b')}$. This gives (ii). Over the open subset of $B \times \mathbb{P}^1$ where π is étale and $u_0 \cdots u_4 \neq 0$, the morphism $\mathcal{C}'' \rightarrow B \times \mathbb{P}^1$ is étale, so for a local coordinate z on B the pair (z, s) is a system of local coordinates on S there, and $q = [x(s, \pi(z))]$ with x as in (3.3). Hence the vector field $\tilde{v}$ which in these coordinates has s -component 0 and constant z -component equal to that of v' is a local holomorphic lift of v' with $dq(\tilde{v}) = \partial x / \partial (d\pi(v'))$. Near the other points of $S_{b'}$ holomorphic lifts exist because p is a submersion there, and two lifts differ by a vertical vector field, whose image under dq is tangent to the fibre. This gives (iii). $\square$

Proposition 3.15. *Let $y^* \in \mathcal{P}$ be a nondegenerate point such that $J_b(y^*)$ is invertible and $\mathcal{I}(y^*, v(y^*))$ is nonzero. Let B, S, p, q be as in Lemma 3.14 and let $Z = (p, q)_*[S]$, a 2-dimensional cycle on $B \times X$. Then $Z^*(\omega_\alpha) \neq 0$ in $H^1(B, \mathbb{C})$ for every nonzero $\omega_\alpha \in V(\alpha)$.*

Proof. Let ω be the harmonic representative of ω_α ; it is a closed form of type $(2, 1)$. We normalize fibre integration by $\int_B p_* \eta \wedge \psi = \int_S \eta \wedge p^* \psi$. Over B° the map p is a proper submersion, so $\vartheta = p_* q^* \omega$ is a smooth closed 1-form on B° ; it has type $(1, 0)$ because $q^* \omega$ has type $(2, 1)$ and the fibres are complex curves, hence it is holomorphic. For $b' \in B^\circ$ and $v' \in T_{b'} B$ one has $\vartheta(v') = \pm \int_{S_{b'}} \iota_{\tilde{v}} q^* \omega$ for any smooth lift $\tilde{v}$ of v' along $S_{b'}$. The integrand does not change if $\tilde{v}$ is changed by a vertical vector field, so $\vartheta(v') = \pm \int_{S_{b'}} \nu^*(\iota_w \omega)$, where w is the first-order deformation of Lemma 3.14(iii).

The form ϑ extends to a holomorphic 1-form on B . Indeed, let $b_0 \in B \setminus B^\circ$ and let z be a coordinate centred at b_0 on a disc D with $D \cap (B \setminus B^\circ) = \{b_0\}$. On $D \setminus \{b_0\}$ the 2-form $\vartheta \wedge d\bar{z}$ is the fibre integral $p_* \Xi$ of the smooth 4-form $\Xi = q^* \omega \wedge p^* d\bar{z}$ on the relatively compact open set $p^{-1}(D) \subset S$, and $|p_* \Xi| \leq p_* |\Xi|$ with $\int_{p^{-1}(D)} |\Xi| < \infty$, so the coefficient of ϑ is integrable on D ; and $\oint_{|z|=r} \vartheta = \pm \int_{p^{-1}\{|z|=r\}} q^* \omega = \pm \int_{p^{-1}\{|z| \leq r\}} q^* d\omega = 0$ for small $r > 0$. A holomorphic 1-form on a punctured disc with integrable coefficient has at most a simple pole, and here the residue vanishes, so ϑ extends holomorphically over b_0 . Since $p^{-1}(B \setminus B^\circ)$ has measure zero, $\int_B \vartheta \wedge \psi = \int_S q^* \omega \wedge p^* \psi = \int_B Z^*(\omega_\alpha) \cup [\psi]$ for every closed 1-form ψ on B , so ϑ represents $Z^*(\omega_\alpha)$.

The set $\mathcal{P}_1^\circ$ is connected, being a nonempty Zariski-open subset of the irreducible curve $\mathcal{P}_1$, and the function $y \mapsto \mathcal{I}(y, v(y))$ is holomorphic on it and nonzero at y^* ; so its zeros form a countable set. As $\pi(B^\circ)$ is uncountable, we can choose $b' \in B^\circ$ with $\mathcal{I}(\pi(b'), v(\pi(b')))) \neq 0$, and $v' \in T_{b'} B$ with $v' \neq 0$. Since π is étale at b' , $d\pi(v') = \mu v(\pi(b'))$ with $\mu \neq 0$, so $\mathcal{I}(\pi(b'), d\pi(v')) \neq 0$. By Lemma 3.14(ii), (iii) and Proposition 3.11, $\lambda \neq 0$ and $\vartheta(v') \neq 0$. A nonzero holomorphic 1-form on a compact Riemann surface is not exact, so $Z^*(\omega_\alpha) = [\vartheta] \neq 0$. $\square$

3.7. From one nonzero value to coniveau one.

Theorem 3.16. *Let B be a smooth connected projective curve and Z a 2-dimensional cycle on $B \times X$ with $Z^*(\omega_\alpha) \neq 0$ for some $\omega_\alpha \in V(\alpha)$. Let $B^G = G \times B$ ($|G|$ disjoint copies of B) and let Z^G be the cycle on $B^G \times X$ whose restriction to $\{g\} \times B \times X$ is $(\text{id}_B \times g)_* Z$. Then $\Lambda = Z_*^G \circ (Z^G)^*$ is an endomorphism of $H^3(X, \mathbb{Q})$ induced by an algebraic cycle on $X \times X$ which restricts to an automorphism of M_α , and there is a polynomial ℓ with rational coefficients such that $\ell(\Lambda)\Lambda$ is the identity on M_α . Consequently*

$$M_\alpha \subset Z_*^G H^1(B^G, \mathbb{Q}) \subset N^1 H^3(X, \mathbb{Q}).$$

Proof. On the copy indexed by g the correspondence Z^G acts as $g_* Z_*$ and its transpose as $Z^* g^*$, so $\Lambda = \sum_{g \in G} g_* Z_* Z^* g^*$. Since G is abelian and $g_* = (g^{-1})^*$, Λ commutes with the action of G . It therefore preserves each eigenspace $V(t\alpha)$, which is one-dimensional, and acts on it by a scalar Λ_t . As Λ is defined over $\mathbb{Q}$, the argument of Lemma 2.3(a) shows that $\Lambda_1 \in \mathbb{Q}(\zeta_{35})$ and that Λ_t is the image of Λ_1 under the automorphism $\zeta_{35} \mapsto \zeta_{35}^t$. It suffices to show $\Lambda_1 \neq 0$: then $\Lambda|_{M_\alpha}$ is invertible, and by the Cayley–Hamilton theorem its inverse is $\ell(\Lambda)|_{M_\alpha}$ for a polynomial ℓ with rational coefficients.

Let $\omega_\alpha \in V(\alpha)$ with $\vartheta := Z^*(\omega_\alpha) \neq 0$; ϑ is a holomorphic 1-form on B because Z^* has bidegree $(-1, -1)$ and $V(\alpha) \subset H^{2,1}$. Using (2.1), $\int_X g_* x \cup x' = \int_X x \cup g^* x'$, $g^* \omega_\alpha = \alpha(g) \omega_\alpha$, $g^* \overline{\omega_\alpha} = \overline{\alpha(g)} \overline{\omega_\alpha}$ and $Z^*(\overline{\omega_\alpha}) = \overline{Z^*(\omega_\alpha)} = \overline{\vartheta}$ (as Z^* is defined over $\mathbb{Q}$), we get

$$\Lambda_1 \int_X \omega_\alpha \cup \overline{\omega_\alpha} = \int_X \Lambda(\omega_\alpha) \cup \overline{\omega_\alpha} = \sum_{g \in G} |\alpha(g)|^2 \int_X Z_* Z^* \omega_\alpha \cup \overline{\omega_\alpha} = |G| \int_B \vartheta \wedge \overline{\vartheta} \neq 0,$$

since $i \int_B \vartheta \wedge \overline{\vartheta} > 0$. Hence $\Lambda_1 \neq 0$. Finally $M_\alpha = \Lambda(M_\alpha) \subset Z_*^G H^1(B^G, \mathbb{Q})$, and Lemma 2.8 gives the last inclusion. $\square$

Proposition 3.17. *There is a nondegenerate point $y^* \in \mathcal{P}$ such that $J_b(y^*)$ is invertible and $\mathcal{I}(y^*, v(y^*))$ is nonzero.*

This is proved by a computer-verified certificate in Appendix A.

Proof of Theorem 1.1(A). Lemma 3.1 gives the assertions on M_α . By Propositions 3.17 and 3.15 there are B and Z such that $Z^*(\omega_\alpha)$ is nonzero, and Theorem 3.16 gives

$$M_\alpha \subset Z_*^G H^1(B^G, \mathbb{Q}) \subset N^1 H^3(X, \mathbb{Q}).$$

Take $C = B^G$ and replace Z by Z^G . $\square$

4. THE HODGE CONJECTURE FOR M_β ON X_{35}^6

Let $X^1 = X_{35}^1$ be the Fermat curve of degree 35 and $\kappa = (17, 22, 31) \in \mathfrak{A}_{35}^1$, so that $\beta = \alpha * \kappa$.

Lemma 4.1. *For every $t \in (\mathbb{Z}/35)^\times$ one has $|t\alpha| + |t\kappa| = 4$; more precisely $|t\kappa| = 2$ for $t \in \mathcal{T}$ and $|t\kappa| = 1$ for $t \notin \mathcal{T}$. Consequently $\beta \in \mathfrak{B}_{35}^6$, and for every t the line $V(t\alpha) \otimes V(t\kappa) \subset H^4(X \times X^1, \mathbb{C})$ is of type $(2, 2)$.*

Proof. The first assertion is read off from Table 2. Then $|t\beta| = |t\alpha| + |t\kappa| = 4 = 6/2 + 1$. By Theorem 2.1(ii), $V(t\kappa) \subset H^{0,1}(X^1)$ for $t \in \mathcal{T}$ and $V(t\kappa) \subset H^{1,0}(X^1)$ otherwise, while $V(t\alpha)$ is of type $(2, 1)$, resp. $(1, 2)$, by Lemma 3.1; so $V(t\alpha) \otimes V(t\kappa)$ has type $(2, 2)$ in both cases. $\square$

In other words, the CM types of M_α and of $M_\kappa \subset H^1(X^1, \mathbb{Q})$ match, so that $M_\alpha \otimes M_\kappa$ contains the 24-dimensional space of Hodge classes $\bigoplus_t V(t\alpha) \otimes V(t\kappa)$.

Proposition 4.2. *Every Hodge class in $M_\alpha \otimes H^1(X^1, \mathbb{Q}) \subset H^4(X \times X^1, \mathbb{Q})$ is algebraic.*

Proof. Let B^G , Z^G , Λ and ℓ be as in Theorem 3.16 (which applies by Proposition 3.15 and Proposition 3.17). Consider the correspondence $Z' = Z^G \times \Delta_{X^1}$ from $B^G \times X^1$ to $X \times X^1$, where Δ_{X^1} is the diagonal. On $H^3(X) \otimes H^1(X^1)$ one has $Z'_* Z'^* = \Lambda \otimes \text{id}$, and $\ell(\Lambda) \otimes \text{id}$ is a morphism of Hodge structures preserving $M_\alpha \otimes H^1(X^1, \mathbb{Q})$. Let v be a Hodge class in $M_\alpha \otimes H^1(X^1, \mathbb{Q})$. Then $v' = Z'^*(\ell(\Lambda) \otimes \text{id})v$ is a Hodge class in $H^2(B^G \times X^1, \mathbb{Q})$, hence algebraic by the Lefschetz (1, 1)-theorem applied to each component of the surface $B^G \times X^1$ [11, p. 163], and $v = (\Lambda \ell(\Lambda) \otimes \text{id})v = Z'_* v'$ is algebraic. $\square$

Theorem 4.3. $\beta \in \mathfrak{C}_{35}^6$. *Equivalently, every element of the 24-dimensional space $M_\beta \subset H^6(X_{35}^6, \mathbb{Q})$ is algebraic.*

Proof. The set of characters $\{(t\alpha, t\kappa)\}$ of $G \times G_{35}^1$ is stable under $\text{Gal}(\mathbb{Q}(\zeta_{35})/\mathbb{Q})$, so $L = \bigoplus_t V(t\alpha) \otimes V(t\kappa)$ is defined over $\mathbb{Q}$: $L = L_{\mathbb{Q}} \otimes \mathbb{C}$ with $L_{\mathbb{Q}} \subset M_\alpha \otimes H^1(X^1, \mathbb{Q})$. By Lemma 4.1, $L_{\mathbb{Q}}$ consists of Hodge classes, which are algebraic by Proposition 4.2. In particular $V(\alpha) \otimes V(\kappa) \subset \text{Alg}^2(X \times X^1) \otimes \mathbb{C}$. Theorem 2.7 with $r - 1 = 3$, $s - 1 = 1$ (so $n = 6$) gives $\alpha * \kappa = \beta \in \mathfrak{C}_{35}^6$. By Lemma 2.3(a), $M_\beta \subset \text{Alg}^3(X_{35}^6)$; and $\dim M_\beta = |\beta| = 24$. $\square$

5. THE GAP GROUP AND ALL DIMENSIONS

5.1. Aoki's groups. Fix $m \geq 2$. Let R_m be the free abelian group on the symbols (y) , $y \in \mathbb{Z}/m \setminus \{0\}$, and for $x = \sum_y c_y(y) \in R_m$ put $\text{len}(x) = \sum_y c_y$. For $a = (a_0, \dots, a_{n+1}) \in \mathfrak{A}_m^n$ put $\mathbf{u}(a) = (a_0) + \dots + (a_{n+1}) \in R_m$. Then $\mathbf{u}(a * b) = \mathbf{u}(a) + \mathbf{u}(b)$, and $\mathbf{u}(a) = \mathbf{u}(b)$ if and only if $a \sim b$. The group $(\mathbb{Z}/m)^\times$ acts on R_m by $t \cdot (y) = (ty)$, and $\mathbf{u}(ta) = t \cdot \mathbf{u}(a)$. For $e \geq 1$ let $e_*: R_m \rightarrow R_{em}$ be the homomorphism $(y) \mapsto (e\langle y \rangle)$, where $\langle y \rangle \in \{1, \dots, m-1\}$ represents y ; thus $\mathbf{u}(ea) = e_* \mathbf{u}(a)$ for the character ea of Lemma 2.3(c).

Following Aoki [3, Section 2] let B_m , D_m and S_m be the subgroups of R_m generated, respectively, by the $\mathbf{u}(a)$ with $a \in \mathfrak{B}_m^n$, by the $\mathbf{u}(d)$ with $d \in \mathfrak{D}_m^n$, and by D_m together with the $\mathbf{u}(\sigma_{p,i})$ for all standard characters $\sigma_{p,i}$ of level m (Section 2.2); here n runs over the even integers ≥ 0 . The group D_m is generated by the elements $(y) + (-y)$. By Lemma 5.2 below, $D_m \subset S_m \subset B_m$. The quotient B_m/S_m is Aoki's *gap group*. In [3, Section 2] the even integer n runs over $n > 0$ only; this gives the same groups, since $(y) + (-y) = \mathbf{u}((y, -y, y, -y, y, -y)) - \mathbf{u}((y, -y, y, -y))$, and since the sets of standard characters agree (Section 2.2).

Lemma 5.1. *An element $x = \sum_y c_y(y)$ of R_m lies in B_m if and only if $\text{len}(x)$ is even and $\sum_y c_y \langle ty \rangle = \frac{m}{2} \text{len}(x)$ for all $t \in (\mathbb{Z}/m)^\times$.*

Proof. Let B'_m be the subgroup of the x satisfying both conditions. If $a \in \mathfrak{B}_m^n$, then $\text{len}(\mathbf{u}(a)) = n + 2$ is even and $\sum_i \langle ta_i \rangle = m|ta| = m(\frac{n}{2} + 1)$ for all t by Theorem 2.1(iii); hence $B_m \subset B'_m$. Conversely, let $x \in B'_m$, and write $x = x_+ - x_-$ with $x_\pm = \sum_y c_y^\pm(y)$ and $c_y^\pm \geq 0$. The element $x_- + \sum_y c_y^-(y) = \sum_y c_y^-(y) + (-y)$ is 0 or $\mathbf{u}(d)$ for some $d \in \mathfrak{D}_m^n \subset \mathfrak{B}_m^n$, and $z = x + \sum_y c_y^-(y) + (-y) = x_+ + \sum_y c_y^-(y)$ lies in B'_m and has nonnegative coefficients. If $z = 0$, then $x \in B_m$. Otherwise $z = \mathbf{u}(a)$, where a lists the symbols occurring in z with their multiplicities. The character a has an even number $n + 2 = \text{len}(z)$ of nonzero entries, $\sum_i \langle a_i \rangle = \frac{m}{2}(n + 2) \equiv 0 \pmod{m}$, and $|ta| = \frac{n}{2} + 1$ for all t ; so $a \in \mathfrak{B}_m^n$, and $x = z - \mathbf{u}(d) \in B_m$. $\square$

Lemma 5.2. *Let p be a prime divisor of m with $m > p$, put $d = m/p$, and let $0 < i < d$.*

- (a) $\sigma_{p,i} \in \mathfrak{B}_m^{p-1}$ if p is odd, and $\sigma_{2,i} \in \mathfrak{B}_m^2$. For $t \in (\mathbb{Z}/m)^\times$ one has $t\sigma_{p,i} \sim \sigma_{p,i'}$, where $0 < i' < d$ and $i' \equiv ti \pmod{d}$; in particular $-\sigma_{p,i} \sim \sigma_{p,d-i}$. For $e \geq 1$ the character $e\sigma_{p,i}$ of level em is $\sigma_{p,ei}$.
- (b) $\sigma_{p,i}$ is algebraic.

Consequently $D_m \subset S_m \subset B_m$, these groups are stable under $(\mathbb{Z}/m)^\times$, and e_* maps D_m , S_m and B_m into D_{em} , S_{em} and B_{em} .

Proof. (a) The first p entries of $\sigma_{p,i}$ (the first two if $p = 2$) are the elements of the coset $i + d\mathbb{Z}/m\mathbb{Z}$, and their representatives $i, i+d, \dots, i+(p-1)d$ lie in $\{1, \dots, m-1\}$. Hence $|\sigma_{p,i}| = (pi + dp(p-1)/2 + (m-pi))/m = (p+1)/2$ for odd p , and $|\sigma_{2,i}| = (i + (i+d) + (m-2i) + d)/m = 2$. Let $t \in (\mathbb{Z}/m)^\times$. Multiplication by t maps the coset $i + d\mathbb{Z}/m\mathbb{Z}$ onto $ti + d\mathbb{Z}/m\mathbb{Z} = i' + d\mathbb{Z}/m\mathbb{Z}$, where $i' \neq 0$ as t is prime to d ; it maps $-pi$ to $-pti = -pi'$, since $p(ti - i') \in pd\mathbb{Z} = m\mathbb{Z}$; and for $p = 2$ it fixes $d = m/2$, since t is odd. Hence $t\sigma_{p,i} \sim \sigma_{p,i'}$ and $|t\sigma_{p,i}| = |\sigma_{p,i'}| = (p+1)/2$, and Theorem 2.1(iii) gives the first assertion. The assertion on $e\sigma_{p,i}$ follows from the definitions.

(b) For $p = 2$ this is Theorem 2.6. Let p be odd, $g = \gcd(i, d)$, $m' = m/g$, $i' = i/g$ and $d' = d/g = m'/p$. Then $0 < i' < d'$, $\gcd(i', d') = 1$, and $\sigma_{p,i} = g\sigma_{p,i'}$ by (a). If $d' \geq 3$, then $\sigma_{p,i'} \in \mathfrak{C}_{m'}^{p-1}$ by Theorem 2.5. If $d' = 2$, then $i' = 1$ and $\sigma_{p,1} = (1, 3, \dots, 2p-1, p)$ at level $2p$ is decomposable: $y \mapsto 2p-y$ pairs the entries $1, 3, \dots, 2p-1$ other than p , and p is paired with the last entry; so $\sigma_{p,1} \in \mathfrak{D}_{2p}^{p-1} \subset \mathfrak{C}_{2p}^{p-1}$ [2, Thm. 1-1]. In both cases $\sigma_{p,i}$ is algebraic by Lemma 2.3(c).

The last assertions follow from (a), since multiplication by t preserves $\mathfrak{B}_m^n$ and $\mathfrak{D}_m^n$, and $a \mapsto ea$ maps $\mathfrak{B}_m^n$ into $\mathfrak{B}_{em}^n$ (Lemma 2.3(c)) and $\mathfrak{D}_m^n$ into $\mathfrak{D}_{em}^n$. $\square$

The following lemma is the step from the gap group to characters in the proof of [3, Thm. 0.1]; it rests on Theorem 2.4, i.e. on [3, Lemma 4.1(i),(ii)].

Lemma 5.3. *Let $\mathcal{A}$ be a set of algebraic characters of level m , each of them in $\mathfrak{B}_m^r$ for some even $r \geq 0$. Let n be even, and let $a \in \mathfrak{B}_m^n$ with $\mathbf{u}(a) \in D_m + \sum_{c \in \mathcal{A}} \mathbb{Z} \mathbf{u}(c)$. Then $a \in \mathfrak{C}_m^n$.*

Proof. Write $\mathbf{u}(a) = \sum_{j=1}^J \epsilon_j \mathbf{u}(c_j) + \sum_y k_y ((y) + (-y))$ with $c_j \in \mathcal{A}$, $\epsilon_j = \pm 1$ and $k_y \in \mathbb{Z}$. For each j with $\epsilon_j = -1$ we use

$$-\mathbf{u}(c_j) = \mathbf{u}(-c_j) - \sum_i ((c_{j,i}) + (-c_{j,i})),$$

where the $c_{j,i}$ are the entries of c_j ; the character $-c_j$ lies in $\mathfrak{B}$ and is algebraic by Lemma 2.3(a) with $t = -1$. This gives $\mathbf{u}(a) = \sum_j \mathbf{u}(c'_j) + \sum_y k'_y ((y) + (-y))$ with algebraic characters $c'_j \in \{c_j, -c_j\}$ of even dimension and integers k'_y . Let d , resp. d' , be the decomposable character consisting of $|k'_y|$ copies of the pair $(y, -y)$ for each y with $k'_y < 0$, resp. $k'_y > 0$ (either may be empty). Then

$$\mathbf{u}(a * d) = \mathbf{u}(a) + \mathbf{u}(d) = \mathbf{u}(c'_1) + \dots + \mathbf{u}(c'_J) + \mathbf{u}(d') = \mathbf{u}(c'_1 * \dots * c'_J * d'),$$

so $a * d \sim c'_1 * \dots * c'_J * d'$; the right side is not empty, since $\text{len}(\mathbf{u}(a * d)) > 0$. It lies in $\mathfrak{C}$ by Theorem 2.4(i), since the c'_j are algebraic and $d' \in \mathfrak{D} \subset \mathfrak{C}$. Hence $a * d \in \mathfrak{C}$ by Lemma 2.3(b), and $a \in \mathfrak{C}_m^n$ by Theorem 2.4(ii) (directly, if d is empty). $\square$

Corollary 5.4. (i) *Let n be even and $a \in \mathfrak{B}_m^n$. Then a is standard if and only if $\mathbf{u}(a) \in S_m$.*
(ii) *Suppose that $B_m = S_m + \sum_{c \in \mathcal{C}} \mathbb{Z} \mathbf{u}(c)$ for a set $\mathcal{C}$ of algebraic characters of level m , each in $\mathfrak{B}_m^r$ for some even r . Then $\mathfrak{B}_m^n = \mathfrak{C}_m^n$ for every even n , and the Hodge conjecture holds for X_m^n for every $n \geq 0$.*

Proof. (i) If $a * d \sim \sigma_1 * \dots * \sigma_k * d'$ as in the definition of standard characters, then $\mathbf{u}(a) = \sum_j \mathbf{u}(\sigma_j) + \mathbf{u}(d') - \mathbf{u}(d) \in S_m$. Conversely, let $\mathbf{u}(a) \in S_m$. The argument of the proof of Lemma 5.3, applied with $\mathcal{A}$ the set of standard characters, which is stable under $c \mapsto -c$ up to the order of the entries by Lemma 5.2(a), gives $a * d \sim c'_1 * \dots * c'_J * d'$ with standard characters c'_j and decomposable d, d' ; so a is standard.

(ii) By definition $S_m = D_m + \sum_\sigma \mathbb{Z} \mathbf{u}(\sigma)$, the sum over the standard characters, and these are algebraic by Lemma 5.2(b). Hence Lemma 5.3, applied with $\mathcal{A}$ the union of $\mathcal{C}$ and the set of standard characters, gives $\mathfrak{B}_m^n = \mathfrak{C}_m^n$ for every even n . Let h be the hyperplane class. For

even n the Hodge classes in $H^n(X_m^n, \mathbb{Q})$ are the sums of a multiple of $h^{n/2}$ and a Hodge class in H_{prim}^n , and the latter span $\bigoplus_{a \in \mathfrak{B}_m^n} V(a)$ over $\mathbb{C}$ by Theorem 2.1(iii). In the degrees $2k \neq n$ one has $H^{2k}(X_m^n, \mathbb{Q}) = \mathbb{Q}h^k$ by the Lefschetz hyperplane theorem and Poincaré duality. $\square$

5.2. Aoki's structure theorem. Let $P = \{4\} \cup \{p : p \text{ an odd prime}\}$. For $q \geq 1$ put $\mu'(q) = (-1)^r$ if q is a product of r distinct elements of P , and $\mu'(q) = 0$ otherwise [3, Section 2]. For $m \geq 3$ let $\mathcal{Q}(m)$ be the set of divisors $q > 1$ of m with $\mu'(q) = 1$. The elements of P are pairwise coprime; hence, if m has exactly r divisors in P , then $\mathcal{Q}(m)$ consists of the products of an even number ≥ 2 of them, and $|\mathcal{Q}(m)| = 2^{r-1} - 1$.

Theorem 5.5 (Aoki, Yamamoto). *Let $m \geq 3$.*

- (i) *For each $q \in \mathcal{Q}(m)$ there is an element $\xi_q \in B_q \setminus S_q$ such that B_m is generated by S_m and the elements $t \cdot (m/q)_* \xi_q$ with $q \in \mathcal{Q}(m)$ and $t \in (\mathbb{Z}/m)^\times$ [3, Thm. 2.1]; see also [1, Thm. D] and the remark below.*
- (ii) *For $x \in B_m$ and $t \in (\mathbb{Z}/m)^\times$ one has $2x \in S_m$ [3, Thm. 2.1], [19], [1, Rem. 5.4], and $x + t \cdot x \in S_m$ [3, proof of Cor. 2.3].*
- (iii) *$B_m/S_m \cong (\mathbb{Z}/2)^{2^{r-1}-1}$, where r is the number of divisors of m in P [28, Thm. 4], [3, Rem. 2.4].*

By (ii), B_m/S_m is a vector space over $\mathbb{F}_2$ on which $(\mathbb{Z}/m)^\times$ acts trivially; by (i) it is spanned by the classes of the $(m/q)_* \xi_q$, $q \in \mathcal{Q}(m)$, and by (iii) these classes form a basis. If q is the product of two elements of P , then $\mathcal{Q}(q) = \{q\}$, so $B_q = S_q + \mathbb{Z}\xi_q$ and $B_q/S_q \cong \mathbb{Z}/2$; every element of $B_q \setminus S_q$ is congruent to ξ_q modulo S_q .

In [1] the notation differs: the group called B_m there is the group $K_m \supset B_m$ of the $x \in R_m$ with $\sum_y c_y \langle ty \rangle = \frac{m}{2} \text{len}(x)$ for all $t \in (\mathbb{Z}/m)^\times$ (see the proof of Proposition 5.8), the group called S_m there is generated by the standard characters alone, and the group called D_m there also contains $(m/2)$ if m is even. So [1, Thm. D] states that K_m is generated by S_m , by $(m/2)$ if m is even, and by the elements $t \cdot (m/q)_* \xi'_q$ for certain $\xi'_q \in K_q$. If $\text{len}(\xi'_q)$ is odd, then q is even (for odd q one has $K_q = B_q$), and we replace ξ'_q by $\xi'_q + (q/2)$, which lies in B_q ; as $(m/q)_*(q/2) = (m/2)$, the generators change only by multiples of $(m/2)$. Comparing lengths, and using $2(m/2) \in D_m$, one obtains the generation statement of (i) with elements $\xi_q \in B_q$. This is all that the proof of Theorem 1.1(C) below uses: if $\xi_q \in S_q$, the corresponding generators are redundant. Likewise, [1, Rem. 5.4] reports Kubert's theorem [19] that $2K_m$ lies in the group generated by S_m and, if m is even, $(m/2)$; for $x \in B_m$, comparing lengths gives $2x \in S_m$.

5.3. Order parities and six representatives.

Lemma 5.6. *Let $\ell \in P$ be a divisor of m with $\gcd(\ell, m/\ell) = 1$, and let $O_\ell \subset \mathbb{Z}/m$ be the set of elements of order ℓ . Let $\nu_\ell: R_m \rightarrow \mathbb{Z}/2$ be the homomorphism $\sum_y c_y(y) \mapsto \sum_{y \in O_\ell} c_y \pmod{2}$, and for a character a write $\nu_\ell(a) = \nu_\ell(\mathbf{u}(a))$, the parity of the number of entries of a of order ℓ . Then ν_ℓ vanishes on S_m . In particular, every $a \in \mathfrak{B}_m^n$ with $\nu_\ell(a) = 1$ is exceptional.*

Proof. As $O_\ell = -O_\ell$, ν_ℓ vanishes on D_m . Let $\sigma = \sigma_{p,i}$ be a standard character, $d = m/p$, and for a divisor k of m let $\Gamma_k \subset \mathbb{Z}/m$ be the subgroup of order k . The entries of σ are the elements of the coset $i + \Gamma_p$, which does not contain 0 as $0 < i < d$, the element $-pi$, and, if $p = 2$, the element d , of order 2. We show that an even number of them lie in O_ℓ ; this proves the first assertion, and the second follows by Corollary 5.4(i).

Suppose that $p \nmid \ell$. Then multiplication by p is an automorphism of Γ_ℓ , and $\Gamma_p \cap \Gamma_\ell = 0$. Hence $i + \Gamma_p$ contains at most one element of Γ_ℓ , and it contains one, say z , if and only if $pi \in \Gamma_\ell$: if $pi = pz$ with $z \in \Gamma_\ell$, then $p(i - z) = 0$, i.e. $i - z \in \Gamma_p$. In that case $-pi = -pz$ has the same

order as z . So the number of entries of σ in O_ℓ is 2 if z exists and lies in O_ℓ , and 0 otherwise (note $\ell \neq 2$).

Suppose that $p = \ell$, an odd prime. Then $i \notin \Gamma_\ell$, so $i + \Gamma_\ell$ does not meet Γ_ℓ ; and $-\ell i \in \Gamma_\ell$ would mean $\ell^2 i = 0$, i.e. m/ℓ divides ℓi , hence i as $\gcd(\ell, m/\ell) = 1$, which is impossible for $0 < i < m/\ell$. So no entry lies in O_ℓ .

Suppose finally that $p = 2$ and $\ell = 4$, so that $8 \nmid m$. The coset $\{i, i + d\} = i + \Gamma_2$ either equals $O_4 = m/4 + \Gamma_2$ or is disjoint from it, d has order 2, and $-2i \in O_4$ would force i to have order 8. So the number is 0 or 2. $\square$

Lemma 5.7. *Put*

$$\begin{aligned} \rho_{12} &= (1, 6, 8, 9), & \rho_{15} &= (1, 6, 10, 13), & \rho_{20} &= (1, 4, 17, 18), \\ \rho_{21} &= (1, 4, 18, 19), & \rho_{28} &= (1, 9, 21, 25), & \rho_{35} &= \beta, \end{aligned}$$

and $(\ell_{12}, \ell_{15}, \ell_{20}, \ell_{21}, \ell_{28}, \ell_{35}) = (3, 5, 5, 7, 4, 5)$. Let $q \in \{12, 15, 20, 21, 28, 35\}$.

- (i) ρ_q is an algebraic character of level q ; $\rho_q \in \mathfrak{B}_q^2$ for $q \neq 35$, and $\rho_{35} \in \mathfrak{B}_{35}^6$.
- (ii) ℓ_q satisfies the hypothesis of Lemma 5.6 for $m = q$, and ρ_q has exactly one entry of order ℓ_q , namely 8, 6, 4, 18, 21, 21, respectively. Hence $\nu_{\ell_q}(\rho_q) = 1$, and $\mathbf{u}(\rho_q) \notin S_q$.
- (iii) $B_q = S_q + \mathbb{Z}\mathbf{u}(\rho_q)$, $B_q/S_q \cong \mathbb{Z}/2$, and $\mathbf{u}(\rho_q) \equiv \xi_q \pmod{S_q}$ for the element ξ_q of Theorem 5.5(i) for $m = q$.

Proof. (i) For $q \neq 35$ one checks $|t\rho_q| = 2$ for all $t \in (\mathbb{Z}/q)^\times$ (a finite computation, see the ancillary file `gap.py`), so $\rho_q \in \mathfrak{B}_q^2$, and ρ_q is algebraic by Theorem 2.6. For $q = 35$ see Theorem 4.3. (ii) The numbers 12, 15, 20, 21, 28, 35 are $3 \cdot 4$, $3 \cdot 5$, $4 \cdot 5$, $3 \cdot 7$, $4 \cdot 7$, $5 \cdot 7$, and ℓ_q is one of the two factors; the entries of order ℓ_q are read off, and Lemma 5.6 applies. (iii) follows from (ii) and the remark after Theorem 5.5, as q is a product of two elements of $\mathbf{P}$. Independently of Theorem 5.5(ii),(iii), the computation described in the proof of Proposition 5.8, carried out for $m = q$, shows $B_q = S_q + \mathbb{Z}\mathbf{u}(\rho_q)$ and $2\mathbf{u}(\rho_q) \in S_q$; together with (ii) this gives $B_q/S_q \cong \mathbb{Z}/2$, and hence $\mathbf{u}(\rho_q) \equiv \xi_q \pmod{S_q}$ for every $\xi_q \in B_q \setminus S_q$. $\square$

The characters $\rho_{12}, \rho_{15}, \rho_{20}$ are the elements $\xi_{12}, \xi_{15}, \xi_{20}$ of [3, p. 185]. For $q = 21$ the element $a = (1, 4, 16, 9, 15, 18) \in \mathfrak{B}_{21}^4$ given there would serve as well: it has $\nu_7(a) = 1$, so $\mathbf{u}(a) \equiv \mathbf{u}(\rho_{21}) \pmod{S_{21}}$ by Lemmas 5.6 and 5.7(iii), and it is algebraic. Indeed, $a = a' * b'$ with $a' = (1, 4, 16)$ and $b' = (9, 15, 18)$ in $\mathfrak{A}_{21}^1$; for every t one has $|ta'|, |tb'| \in \{1, 2\}$ and $|ta'| + |tb'| = |ta| = 3$, so by Theorem 2.1(ii) the lines $V(ta') \otimes V(tb')$ are of type $(1, 1)$; their sum is defined over $\mathbb{Q}$, hence consists of algebraic classes on the surface $X_{21}^1 \times X_{21}^1$ by the Lefschetz $(1, 1)$ -theorem, and Theorem 2.7 with $r - 1 = s - 1 = 1$ gives $a \in \mathfrak{C}_{21}^4$. The element printed there for $q = 28$, $(1, 9, 25, 12, 20, 24)$, is not a character, as its entries add up to $91 \not\equiv 0 \pmod{28}$; we use ρ_{28} instead.

5.4. Ten levels.

Proposition 5.8. *Let $m = 35k$ with $1 \leq k \leq 10$. Then $\mathcal{Q}(m)$ is $\{35\}$ for $k \in \{1, 2, 5, 7, 10\}$, $\{15, 21, 35\}$ for $k \in \{3, 6, 9\}$, and $\{20, 28, 35\}$ for $k \in \{4, 8\}$; in each case it is the set of $q \in \{12, 15, 20, 21, 28, 35\}$ dividing m . For $q \in \mathcal{Q}(m)$ let $\rho_q^{(m)}$ denote the character $(m/q)\rho_q$ of level m (Lemma 2.3(c)). Then:*

- (a) $B_m = S_m + \sum_{q \in \mathcal{Q}(m)} \mathbb{Z}\mathbf{u}(\rho_q^{(m)})$;
- (b) $2\mathbf{u}(\rho_q^{(m)}) \in S_m$ for every $q \in \mathcal{Q}(m)$, and $B_m/S_m \cong (\mathbb{Z}/2)^{|\mathcal{Q}(m)|}$, with basis the classes of the $\mathbf{u}(\rho_q^{(m)})$;
- (c) $\mathbf{u}(\rho_{35}^{(m)}) \notin S_m + \sum_{q \in \mathcal{Q}(m) \setminus \{35\}} \mathbb{Z}\mathbf{u}(\rho_q^{(m)})$.

Proof. The description of $\mathcal{Q}(m)$ is elementary, since the divisors of m in $\mathbf{P}$ are among 3, 4, 5, 7. Statements (a)–(c) are proved by an exact computation with integers, carried out in the ancillary file `gap.py`, as follows.

Let $K_m \subset R_m$ be the subgroup of the x with $\sum_y c_y \langle ty \rangle = \frac{m}{2} \text{len}(x)$ for all $t \in (\mathbb{Z}/m)^\times$. The conditions for t and $-t$ coincide, and the remaining $|(\mathbb{Z}/m)^\times|/2$ linear forms are linearly independent, as their rank modulo the prime $2^{61} - 1$ shows; so K_m has rank $r_m = m - 1 - |(\mathbb{Z}/m)^\times|/2$. By Lemma 5.1, $B_m = \{x \in K_m : \text{len}(x) \text{ even}\}$. Hence $B_m = K_m$ if m is odd, and $[K_m : B_m] = 2$ if m is even, since then $(m/2) \in K_m \setminus B_m$.

(a) Let L_m be the subgroup generated by the elements $(y) + (-y)$, the $\mathbf{u}(\sigma)$ for the standard characters σ of level m , and the $\mathbf{u}(\rho_q^{(m)})$; each of these generators lies in B_m by Lemmas 5.2(a) and 5.7(i), and this is also checked directly with Lemma 5.1. Put $L_m^+ = L_m + \mathbb{Z}(m/2)$ if m is even, and $L_m^+ = L_m$ if m is odd. Unimodular operations on the generators give a basis of L_m^+ in echelon form. It has r_m elements, the product of its pivots is 1 except for $m = 315$, where it is 6, and for $m = 315$ the basis has rank r_m modulo 2 and modulo 3. Hence the maximal minors of this basis have no common prime factor, i.e. L_m^+ is a saturated subgroup of R_m ; as it is contained in K_m and has the same rank, $L_m^+ = K_m$. If m is odd, this is (a). If m is even, then $2(m/2) \in L_m$, so $[K_m : L_m] \leq 2$; since $L_m \subset B_m$ and $[K_m : B_m] = 2$, we get $L_m = B_m$.

(b), (c) Reduction with respect to an echelon basis of S_m shows $2\mathbf{u}(\rho_q^{(m)}) \in S_m$ for $q \in \mathcal{Q}(m)$. For each $q \in \mathcal{Q}(m)$ there are a set $W_q \subset \mathbb{Z}/m \setminus \{0\}$ and $b_q \in \{0, 1\}$, printed explicitly by `gap.py` (as a union of sets of elements of a given order, or, for $m = 280$ and $q = 28$, as a list of 52 residues), such that the homomorphism $\phi_q : B_m \rightarrow \mathbb{Z}/2$, $\phi_q(x) = \sum_{y \in W_q} c_y + b_q \text{len}(x)/2 \pmod{2}$, vanishes on all the generators of S_m listed above and satisfies $\phi_q(\mathbf{u}(\rho_{q'}^{(m)})) = 1$ for $q' = q$ and $= 0$ for $q' \in \mathcal{Q}(m) \setminus \{q\}$. Together with (a), this proves (b) and (c).

As an independent check, the Hermite and Smith normal forms of the generators, computed with FLINT [9] through python-flint, give

$$[K_m : S_m] = 2^{|\mathcal{Q}(m)|} [K_m : B_m], \quad [K_m : L_m] = [K_m : B_m],$$

$$\left[K_m : S_m + \sum_{q \in \mathcal{Q}(m) \setminus \{35\}} \mathbb{Z} \mathbf{u}(\rho_q^{(m)}) \right] = 2 [K_m : B_m].$$

□

Remark 5.9. (a) For $m = 35$ no computation is needed. By Theorem 5.5(iii), $B_{35}/S_{35} \cong \mathbb{Z}/2$, and $\mathbf{u}(\beta) \notin S_{35}$ since $\nu_5(\beta) = 1$ (Lemma 5.7(ii)). Alternatively, let $\varpi(a) \in \mathbb{Z}/2$ be the parity of the number of entries of a character a of level 35 lying in $\{\pm 1, \pm 6\}$. Then ϖ is additive under juxtaposition and vanishes on every decomposable character. The standard characters of level 35 are the $\sigma_{5,i}$ ($1 \leq i \leq 6$) and the $\sigma_{7,i}$ ($1 \leq i \leq 4$); among them only $\sigma_{5,1} = (1, 8, 15, 22, 29, 30)$, $\sigma_{5,6} = (6, 13, 20, 27, 34, 5)$, $\sigma_{7,1} = (1, 6, 11, 16, 21, 26, 31, 28)$ and $\sigma_{7,4} = (4, 9, 14, 19, 24, 29, 34, 7)$ have entries in $\{\pm 1, \pm 6\}$, two each. So ϖ vanishes on S_{35} , while $\varpi(\beta) = 1$. Hence $B_{35} = S_{35} + \mathbb{Z} \mathbf{u}(\beta)$.

(b) Given Theorem 5.5, Proposition 5.8 holds at every level, by the argument in the proof of Theorem 1.1(C) below. The computation above does not use Theorem 5.5, and at the ten levels it confirms Theorem 5.5(i)–(iii); for the assertion $x + t \cdot x \in S_m$ of (ii), `gap.py` also checks $\mathbf{u}(\rho_q^{(m)}) + t \cdot \mathbf{u}(\rho_q^{(m)}) \in S_m$ for all $q \in \mathcal{Q}(m)$ and $t \in (\mathbb{Z}/m)^\times$, which suffices by (a), since S_m is stable under $(\mathbb{Z}/m)^\times$. For $m \neq 280, 315$ the order parities ν_ℓ of Lemma 5.6 with $\ell \in \{3, 4, 5, 7\}$ already separate the classes of the $\mathbf{u}(\rho_q^{(m)})$, so that there Proposition 5.8 also follows by hand from Theorem 5.5(ii),(iii).

5.5. Proof of Theorem 1.1(B) and (C).

Proof of Theorem 1.1(B). Let $m = 35k$ with $1 \leq k \leq 10$. By Lemma 5.7(i) and Lemma 2.3(c), the characters $\rho_q^{(m)}$, $q \in \mathcal{Q}(m)$, are algebraic and lie in $\mathfrak{B}_m^r$ with $r \in \{2, 6\}$. By Proposition 5.8(a) and Corollary 5.4(ii), the Hodge conjecture holds for X_m^n for every n . $\square$

Proof of Theorem 1.1(C). Let $m = 2^a 3^b 5^c 7^d$ with $420 \nmid m$. For $m \leq 2$, X_m^n is a linear subspace or a quadric, whose cohomology is spanned by classes of linear subspaces; so let $m \geq 3$. The divisors of m in $\mathbb{P}$ are among 3, 4, 5, 7, and not all four of them divide m . Hence $\mathcal{Q}(m)$ is the set of $q \in \{12, 15, 20, 21, 28, 35\}$ dividing m . For $q \in \mathcal{Q}(m)$, Lemma 5.7(iii) gives $\xi_q - \mathbf{u}(\rho_q) \in S_q$ (by its proof, this uses from Theorem 5.5 only that $\xi_q \in B_q \setminus S_q$), hence $(m/q)_* \xi_q - \mathbf{u}(\rho_q^{(m)}) \in S_m$ by Lemma 5.2. Since S_m is stable under $(\mathbb{Z}/m)^\times$, Theorem 5.5(i) gives

$$B_m = S_m + \sum_{q \in \mathcal{Q}(m)} \sum_{t \in (\mathbb{Z}/m)^\times} \mathbb{Z} \mathbf{u}(t \rho_q^{(m)}).$$

The characters $t \rho_q^{(m)}$ are algebraic by Lemma 5.7(i) and Lemma 2.3(a),(c), and Corollary 5.4(ii) gives the assertion. $\square$

Remark 5.10. Let m be as in Theorem 1.1(C) with $35 \mid m$, and put

$$L'_m = S_m + \sum_{q \in \mathcal{Q}(m) \setminus \{35\}} \mathbb{Z} \mathbf{u}(\rho_q^{(m)}).$$

By Theorem 5.5 (and, for $m = 35k$ with $k \leq 10$, by Proposition 5.8), L'_m has index 2 in B_m and does not contain $\mathbf{u}(\rho_{35}^{(m)})$. We say that $a \in \mathfrak{B}_m^n$, n even, lies in the class of β if $\mathbf{u}(a) \notin L'_m$. Such characters are exceptional. If $\mathbf{u}(a) \in L'_m$, then a is algebraic by Lemma 5.3, applied with $\mathcal{A}$ the set of the standard characters and the $\rho_q^{(m)}$ with $q \neq 35$; this does not use Theorem 1.1(A). For every even $n \geq 6$ the character $\rho_{35}^{(m)} * d$ with $d \in \mathfrak{D}_m^{n-6}$ lies in the class of β , and if m is even, so does $(m/70)\gamma \in \mathfrak{B}_m^4$ (Remark 7.5). For $m = 35k$ with $k \leq 10$ a finite computation (ancillary file `gap.py`) shows that no character in $\mathfrak{B}_m^2$ lies in the class of β , and, if m is odd, that none in $\mathfrak{B}_m^4$ does. For this, the homomorphism ϕ_{35} of the proof of Proposition 5.8 can be chosen with $b_{35} = 0$ and W_{35} a union of sets $\{y : y \text{ has order } k\}$, so that it is invariant under $(\mathbb{Z}/m)^\times$; it vanishes on all $a \in \mathfrak{B}_m^2$, and for odd m on all $a \in \mathfrak{B}_m^4$, having an entry dividing m , and every orbit of $(\mathbb{Z}/m)^\times$ on $\mathfrak{B}_m^n$ contains such a character (if y is an entry of a and $g = \gcd(y, m)$, then $ty = g$ for some $t \in (\mathbb{Z}/m)^\times$). Thus at the ten levels of Theorem 1.1(B), $\mathfrak{B}_m^n$ contains characters in the class of β exactly for the even $n \geq 6$ if m is odd, and for the even $n \geq 4$ if m is even.

Remark 5.11. Let $m = 2^a 3^b 5^c 7^d$ with $420 \mid m$. Then $\mathcal{Q}(m) = \{12, 15, 20, 21, 28, 35, 420\}$, and by Theorem 5.5 and Lemma 5.7(iii) the subgroup $L'' = S_{420} + \sum_{q \in \mathcal{Q}(420) \setminus \{420\}} \mathbb{Z} \mathbf{u}(\rho_q^{(420)})$ has index 2 in B_{420} . By Lemma 5.1 and its proof there are Hodge characters ρ of level 420 with $\mathbf{u}(\rho) \notin L''$. The proof of Theorem 1.1(C) shows that if $V(\rho)$ is algebraic for one such ρ , then the Hodge conjecture holds for X_m^n for every n and every $m = 2^a 3^b 5^c 7^d$. We do not know an algebraic character of this kind. By Theorem 8.1, such characters have dimension at least 14, and there are such characters of dimension 14.

6. PRODUCTS AND ABELIAN VARIETIES

Corollary 6.1. *Let m be as in Theorem 1.1(C). For all $k \geq 1$ and $r_1, \dots, r_k \geq 0$ the Hodge conjecture holds for $X_m^{r_1} \times \dots \times X_m^{r_k}$.*

Proof. By [24, Thm. IV], the Hodge conjecture for this product follows from the Hodge conjecture for X_m^r for all even $r \leq r_1 + \dots + r_k + 2(k-1)$, which holds by Theorem 1.1(C). $\square$

Following [3], an abelian variety is of *Fermat type of degree m* if it is isogenous to a factor of a power of the Jacobian J_m of the Fermat curve X_m^1 .

Corollary 6.2. *Let m be as in Theorem 1.1(C). Then the Hodge conjecture holds for every abelian variety of Fermat type of degree m . In particular, it holds for every power of the Jacobian of the curve $C_{35}: y^2 = x^{35} - 1$.*

Proof. By [3, Cor. 3.2], the Hodge conjecture for X_m^n for all n implies the Hodge conjecture for all abelian varieties of Fermat type of degree m ; so the first assertion follows from Theorem 1.1(C). The curve C_{35} is a quotient of X_{35}^1 [3, p. 178]: the map $(u, v) \mapsto (x, w) = (u^{35}, uv)$ sends the affine curve $u^{35} + v^{35} = 1$ onto $w^{35} = x(1 - x)$, and $(x, w) \mapsto (2i(x - \frac{1}{2}), 4^{1/35}w)$ maps the latter curve birationally onto $y^2 = z^{35} - 1$. Hence the Jacobian of C_{35} and all its powers are of Fermat type of degree 35. $\square$

The Hodge conjecture for the Jacobian of C_{35} was left open in [4, Example 8.6].

7. TWO CLASSES ON FERMAT FOURFOLDS

By Theorem 1.1(B) the Hodge conjecture holds for X_{70}^4 and X_{210}^4 . In this section we derive the algebraicity of the Hodge structures $M_\gamma \subset H^4(X_{70}^4, \mathbb{Q})$ and $M_\delta \subset H^4(X_{210}^4, \mathbb{Q})$, where

$$\gamma = (1, 20, 24, 42, 61, 62), \quad \delta = (2, 9, 129, 142, 168, 180),$$

directly from Theorem 4.3 by explicit identities, without Proposition 5.8 (see Remark 1.2(d)).

7.1. The class γ . Put

$$\begin{aligned} 2\beta &= (2, 4, 32, 42, 60, 34, 44, 62) \in \mathfrak{A}_{70}^6, & \varsigma_1 &= (1, 18, 53, 68), & \varsigma_2 &= (17, 26, 36, 61), \\ \sigma_{5,5} &= (5, 12, 19, 26, 33, 10) \in \mathfrak{A}_{35}^4, & \sigma_3 &= 2\sigma_{5,5} = (10, 24, 38, 52, 66, 20) \in \mathfrak{A}_{70}^4, \\ \varepsilon &= (2, 68, 4, 66, 10, 60, 17, 53, 18, 52, 26, 44, 32, 38, 34, 36) \in \mathfrak{A}_{70}^{14}. \end{aligned}$$

Here $\sigma_{5,5}$ is Aoki's standard character for $m = 35$, $p = 5$, $d = 7$, $i = 5$, and 2β , σ_3 are as in Lemma 2.3(c) with $e = 2$. The characters ς_1, ς_2 are the surface classes $(i, 35 + i, 35 + 2i, 70 - 4i)$ with $i = 18$, resp. $i = 26$, of the shape of [1, Thm. C(ii)] for $m = 70$ (that theorem is stated for $m > 180$); they are standard in the sense of Section 2.2, since $\varsigma_1 * (36, 34, 35, 35) \sim \sigma_{2,18} * \sigma_{2,1}$ and $\varsigma_2 * (52, 18, 35, 35) \sim \sigma_{2,26} * \sigma_{2,17}$ at level 70. Similarly $\sigma_3 = \sigma_{5,10}$ at level 70 up to the order of the entries; as $\gcd(10, 14) = 2$, Theorem 2.5 does not apply to it directly, and we obtain $\sigma_3 \in \mathfrak{C}_{70}^4$ from level 35.

Proposition 7.1. $\gamma \in \mathfrak{B}_{70}^4$, $\varsigma_1, \varsigma_2 \in \mathfrak{B}_{70}^2$, $\varepsilon \in \mathfrak{D}_{70}^{14}$, and, up to the order of the entries,

$$\gamma * \varepsilon \sim 2\beta * \varsigma_1 * \varsigma_2 * \sigma_3.$$

Proof. The membership statements are finite computations with Theorem 2.1(iii); ε consists of the eight pairs $\{2, 68\}, \{4, 66\}, \{10, 60\}, \{17, 53\}, \{18, 52\}, \{26, 44\}, \{32, 38\}, \{34, 36\}$. Both sides of the identity are the multiset listed in (B.1) of Appendix B. $\square$

Proposition 7.2. $\gamma \in \mathfrak{C}_{70}^4$. Equivalently, every element of the 24-dimensional space $M_\gamma \subset H^4(X_{70}^4, \mathbb{Q})$ is algebraic.

Proof. By Theorem 4.3 and Lemma 2.3(c), $2\beta \in \mathfrak{C}_{70}^6$. By Theorem 2.6, $\varsigma_1, \varsigma_2 \in \mathfrak{C}_{70}^2$. By Theorem 2.5 ($p = 5$, $m = 35$, $d = 7$, $\gcd(5, 7) = 1$), $\sigma_{5,5} \in \mathfrak{C}_{35}^4$, hence $\sigma_3 \in \mathfrak{C}_{70}^4$ by Lemma 2.3(c). Three applications of Theorem 2.4(i) give $2\beta * \varsigma_1 \in \mathfrak{C}_{70}^{10}$, $2\beta * \varsigma_1 * \varsigma_2 \in \mathfrak{C}_{70}^{14}$ and $2\beta * \varsigma_1 * \varsigma_2 * \sigma_3 \in \mathfrak{C}_{70}^{20}$. By Proposition 7.1 and Lemma 2.3(b), $\gamma * \varepsilon \in \mathfrak{C}_{70}^{20}$, and Theorem 2.4(ii) with $r = 4$, $s = 14$ gives $\gamma \in \mathfrak{C}_{70}^4$. Finally Lemma 2.3(a) applies, and $|\gamma| = 24$. $\square$

7.2. The class δ . Put

$$\begin{aligned} 3\gamma &= (3, 60, 72, 126, 183, 186), & \tau &= (24, 94, 138, 164), \\ 47\delta &= (94, 3, 183, 164, 126, 60), & \varepsilon' &= (24, 186, 72, 138), \end{aligned}$$

characters of G_{210}^4 , G_{210}^2 , G_{210}^4 and G_{210}^2 ; here 3γ is as in Lemma 2.3(c) with $e = 3$. The following identity is the exchange identity of [17, Prop. 4.2] (where $-\tau$ is written $q = (46, 72, 116, 186)$), in the form of [17, Thm. 4.1].

Proposition 7.3 (Jumagulov). $\delta \in \mathfrak{B}_{210}^4$, $\tau \in \mathfrak{B}_{210}^2$, $\varepsilon' \in \mathfrak{D}_{210}^2$, and, up to the order of the entries,

$$3\gamma * \tau \sim 47\delta * \varepsilon'.$$

Proof. Both sides are the multiset $\{3, 24, 60, 72, 94, 126, 138, 164, 183, 186\}$; ε' consists of the pairs $\{24, 186\}$ and $\{72, 138\}$. The memberships are finite computations with Theorem 2.1(iii). $\square$

Proposition 7.4. $\delta \in \mathfrak{C}_{210}^4$. Equivalently, every element of the 48-dimensional space $M_\delta \subset H^4(X_{210}^4, \mathbb{Q})$ is algebraic.

Proof. By Proposition 7.2 and Lemma 2.3(c), $3\gamma \in \mathfrak{C}_{210}^4$; by Theorem 2.6, $\tau \in \mathfrak{C}_{210}^2$. Theorem 2.4(i) gives $3\gamma * \tau \in \mathfrak{C}_{210}^8$, hence $47\delta * \varepsilon' \in \mathfrak{C}_{210}^8$ by Proposition 7.3 and Lemma 2.3(b), and Theorem 2.4(ii) with $r = 4$, $s = 2$ gives $47\delta \in \mathfrak{C}_{210}^4$. Since $\gcd(47, 210) = 1$, Lemma 2.3(a) gives $\delta \in \mathfrak{C}_{210}^4$ and $M_\delta \subset \text{Alg}^2(X_{210}^4)$. The 48 characters $t\delta$ are pairwise distinct (the entries 2 and 9 force $t \equiv 1 \pmod{105}$ and $\pmod{70}$), so $\dim M_\delta = 48$. $\square$

Remark 7.5. The identities above show that γ and δ lie in the class of β in the sense of Remark 5.10. Indeed, ς_1 , ς_2 and $\sigma_3 = \sigma_{5,10}$ are standard, so Proposition 7.1 gives $\mathbf{u}(\gamma) - \mathbf{u}(2\beta) \in S_{70}$. As $\tau = \sigma_{3,24}$ at level 210, Proposition 7.3 gives $\mathbf{u}(47\delta) - \mathbf{u}(3\gamma) \in S_{210}$, and $\mathbf{u}(3\gamma) - \mathbf{u}(6\beta) = 3_*(\mathbf{u}(\gamma) - \mathbf{u}(2\beta)) \in S_{210}$ by Lemma 5.2; by Theorem 5.5(ii), $\mathbf{u}(\delta) \equiv \mathbf{u}(47\delta) \pmod{S_{210}}$, so that $\mathbf{u}(\delta) - \mathbf{u}(6\beta) \in S_{210}$ (this is also checked directly in `gap.py`). For every even m with $35 \mid m$ it follows that $\mathbf{u}((m/70)\gamma) - \mathbf{u}((m/35)\beta) \in S_m$. The exceptionality of β , γ and δ can also be read off from Lemma 5.6: their entries of order 5 are 21, 42 and 168, one each, so $\nu_5(\beta) = \nu_5(\gamma) = \nu_5(\delta) = 1$.

8. THE LEVEL 420

Throughout this section $m = 420$. The divisors of 420 in $\mathbf{P}$ are 3, 4, 5, 7, so that $\mathcal{Q}(420) = \{12, 15, 20, 21, 28, 35, 420\}$, and we put

$$L''_{420} = S_{420} + \sum_{q \in \mathcal{Q}(420) \setminus \{420\}} \mathbb{Z} \mathbf{u}(\rho_q^{(420)}), \quad \rho_q^{(420)} = (420/q)\rho_q,$$

as in Proposition 5.8 and Remark 5.11. The characters $\rho_q^{(420)}$ are algebraic by Lemma 5.7(i) and Lemma 2.3(c) (for $q = 35$ this uses Theorem 4.3). Hence, by Lemma 5.3 applied with $\mathcal{A}$ the set of the standard characters and the $\rho_q^{(420)}$, $q \neq 420$, every $a \in \mathfrak{B}_{420}^n$ with $\mathbf{u}(a) \in L''_{420}$ is algebraic.

For $x = \sum_y c_y(y) \in R_{420}$ and a divisor $N > 1$ of 420 let $n_N(x) = \sum_{\text{ord}(y)=N} c_y$, where $\text{ord}(y)$ is the order of y in $\mathbb{Z}/420$; for a character a we write $n_N(a) = n_N(\mathbf{u}(a))$, the number of entries of a of order N . Put $\nu_{15}(x) = n_{15}(x) \pmod{2}$,

$$\mathcal{R} = \{2, 3, 5, 6, 7, 10, 14, 21, 30, 42, 70, 210\}, \quad n_{\mathcal{R}}(x) = \sum_{N \in \mathcal{R}} n_N(x),$$

and let $n'_{420}(x) = \sum_y c_y$, the sum over the units $y \in (\mathbb{Z}/420)^\times$ with $y \equiv \pm 2 \pmod{5}$. Finally let

$$\begin{aligned} w_1 &= (56, 69, 176, 191, 193, 199, 204, 209, 210, 236, 243, 253, 257, 267, 296, 301), \\ w_2 &= (1, 17, 28, 132, 141, 168, 191, 199, 236, 253, 257, 267, 296, 371, 387, 416). \end{aligned}$$

- Theorem 8.1.** (i) L''_{420} has index 2 in B_{420} , and $L''_{420} = \{x \in B_{420} : \nu_{15}(x) = 0\}$.
(ii) Let $a \in \mathfrak{B}_{420}^n$ with $\mathbf{u}(a) \notin L''_{420}$. Then $n_{420}(a) \geq 6$, $n_{105}(a) \geq 3$, $n_{140}(a) \geq 3$ and $n_{15}(a), n_{35}(a), n_{60}(a), n_{\mathcal{R}}(a) \geq 1$. In particular a has at least 16 entries, i.e. $n \geq 14$.
(iii) $w_1, w_2 \in \mathfrak{B}_{420}^{14}$, and $\mathbf{u}(w_1), \mathbf{u}(w_2) \notin L''_{420}$. Each of them has exactly 6, 1, 1, 1, 3, 3 entries of order 420, 15, 35, 60, 105, 140, respectively, and one entry with order in $\mathcal{R}$; so equality holds in (ii). The characters w_1 and w_2 are not Galois conjugate, and $\dim M_{w_1} = \dim M_{w_2} = 96$.
(iv) $\mathfrak{C}_{420}^n = \mathfrak{B}_{420}^n$ for every even $n \leq 12$. Hence the Hodge conjecture holds for X_{420}^n for every $n \leq 12$, and for every product $X_{420}^{r_1} \times \cdots \times X_{420}^{r_k}$ with $r_1 + \cdots + r_k + 2(k-1) \leq 12$.

Part (i) is Proposition 8.3 below, and (ii) is proved in Section 8.4 from four lemmas on character sums (Lemmas 8.5–8.8).

8.1. The class of ξ_{420} .

Lemma 8.2. Let $m \geq 3$, and let ℓ be an odd squarefree divisor of m with $\gcd(\ell, m/\ell) = 1$. Let $O_\ell \subset \mathbb{Z}/m$ be the set of elements of order ℓ and $\nu_\ell: R_m \rightarrow \mathbb{Z}/2$ the homomorphism $\sum_y c_y(y) \mapsto \sum_{y \in O_\ell} c_y \bmod 2$. Then ν_ℓ vanishes on S_m .

Proof. As $O_\ell = -O_\ell$, ν_ℓ vanishes on D_m . Let $\sigma = \sigma_{p,i}$ be a standard character, $d = m/p$, and let $\Gamma_k \subset \mathbb{Z}/m$ denote the subgroup of order k . If $p \nmid \ell$, the argument in the proof of Lemma 5.6 for this case applies without change (it uses only that $p \nmid \ell$ and $\ell \neq 2$) and shows that 0 or 2 entries of σ lie in O_ℓ . Let $p \mid \ell$. Then p is odd, and $p^2 \nmid m$ because ℓ is squarefree and $\gcd(\ell, m/\ell) = 1$; so $\mathbb{Z}/m = \Gamma_p \oplus \Gamma_{m/p}$, where p does not divide the orders of the elements of $\Gamma_{m/p}$. Write $i = i_p + i'$ accordingly. Then $i' \neq 0$, since $i \notin \Gamma_p = d\mathbb{Z}/m\mathbb{Z}$ as $0 < i < d$. The first p entries of σ are the elements $z + i'$, $z \in \Gamma_p$, of the coset $i + \Gamma_p$, and $z + i'$ has order $\text{ord}(z) \text{ord}(i')$; this order is ℓ if and only if $z \neq 0$ and $\text{ord}(i') = \ell/p$. So either 0 or $p-1$ of these entries lie in O_ℓ . The last entry $-pi = -pi'$ lies in $\Gamma_{m/p}$, so its order is prime to p and it does not lie in O_ℓ . In both cases the number of entries of σ in O_ℓ is even. $\square$

Proposition 8.3. (a) ν_{15} vanishes on L''_{420} .

- (b) $w_1, w_2 \in \mathfrak{B}_{420}^{14}$, and each of them has exactly one entry of order 15, namely 56, resp. 28; so $\nu_{15}(w_1) = \nu_{15}(w_2) = 1$.
(c) $[B_{420} : L''_{420}] = 2$, and $L''_{420} = \{x \in B_{420} : \nu_{15}(x) = 0\}$.

Proof. (a) Lemma 8.2 with $\ell = 15$ applies, as $\gcd(15, 28) = 1$; so ν_{15} vanishes on S_{420} . An entry of $\rho_q^{(420)} = (420/q)\rho_q$ of order 15 comes from an entry of ρ_q of order 15 in $\mathbb{Z}/q$. Hence $\rho_q^{(420)}$ has no such entry for $q \in \{12, 20, 21, 28, 35\}$, and $\rho_{15}^{(420)} = (28, 168, 280, 364)$ has two, 28 and 364.

(b) The membership is a finite computation with Theorem 2.1(iii) (ancillary file `level420.py`); the entries of order 15 are read off.

(c) By Theorem 5.5(i), B_{420} is generated by S_{420} and the elements $t \cdot (420/q)_* \xi_q$, $q \in Q(420)$, $t \in (\mathbb{Z}/420)^\times$. For $q \neq 420$, $(420/q)_* \xi_q \equiv \mathbf{u}(\rho_q^{(420)}) \pmod{S_{420}}$ by Lemma 5.7(iii) and Lemma 5.2, and $t \cdot x \equiv -x \equiv x \pmod{S_{420}}$ for $x \in B_{420}$ by Theorem 5.5(ii). Hence $B_{420} = L''_{420} + \mathbb{Z} \xi_{420}$ with $2\xi_{420} \in S_{420}$, and $[B_{420} : L''_{420}] \leq 2$. By (a) and (b), ν_{15} is a surjective homomorphism $B_{420} \rightarrow \mathbb{Z}/2$ vanishing on L''_{420} ; so $[B_{420} : L''_{420}] = 2$ and L''_{420} is its kernel.

Independently of Theorem 5.5, `level420.py` shows $B_{420} = L''_{420} + \mathbb{Z} \mathbf{u}(w_1)$ and $2\mathbf{u}(w_1) \in S_{420}$, by the computation described in the proof of Proposition 5.8(a) (the echelon basis of $L''_{420} + \mathbb{Z} \mathbf{u}(w_1) + \mathbb{Z}(210)$ has $r_{420} = 371$ rows, the product of its pivots is 4, and it has rank 371 modulo 2); together with (a) and (b) this proves (c) as well. The Smith normal forms computed with FLINT give $[K_{420} : S_{420}] = 256$ and $[K_{420} : L''_{420}] = 4$, in accordance with $B_{420}/S_{420} \cong (\mathbb{Z}/2)^7$ (Theorem 5.5(iii)). $\square$

8.2. Character sums. Let $m \geq 3$. For a divisor $N > 1$ of m , every $y \in \mathbb{Z}/m$ of order N can be written uniquely as $y = (m/N)u_y$ with $u_y \in (\mathbb{Z}/N)^\times$. For a Dirichlet character χ of conductor f dividing N we write $\chi(u_y)$ for the value of χ at $u_y \bmod f$, and for $x = \sum_y c_y(y) \in R_m$ we put

$$s_N(\chi; x) = \sum_{\text{ord}(y)=N} c_y \chi(u_y).$$

Let $B_{1,\chi} = \frac{1}{f} \sum_{a=1}^f \chi(a) a$ be the first generalized Bernoulli number. If χ is odd, then $B_{1,\chi} \neq 0$ [27, Chapter 4]; for the characters used below this is also checked directly in `level420.py`.

Lemma 8.4. *Let χ be an odd primitive Dirichlet character whose conductor f divides m , and let $x \in K_m$ (for instance $x \in B_m$). Then*

$$(8.1) \quad \sum_{f|N|m} \frac{\varphi(m)}{\varphi(N)} \prod_{p|N, p \nmid f} (1 - \bar{\chi}(p)) s_N(\chi; x) = 0,$$

where φ is Euler's function and p runs over the primes dividing N but not f .

Proof. For $t \in (\mathbb{Z}/m)^\times$ put $g(t) = \sum_y c_y (2\langle ty \rangle - m)$; as $x \in K_m$, $g = 0$. Hence $\sum_t \bar{\chi}(t) g(t) = \sum_y c_y G(y) = 0$, where $G(y) = \sum_{t \in (\mathbb{Z}/m)^\times} \bar{\chi}(t) (2\langle ty \rangle - m)$ and $\bar{\chi}$ is regarded as a character modulo m . Let y have order N . Then $2\langle ty \rangle - m = \frac{m}{N} (2\langle tu_y \rangle_N - N)$, where $\langle \cdot \rangle_N$ denotes the representative in $\{1, \dots, N-1\}$, and this depends only on $t \bmod N$. If $f \nmid N$, then $\bar{\chi}$ is nontrivial on the kernel of $(\mathbb{Z}/m)^\times \rightarrow (\mathbb{Z}/N)^\times$, since f is the conductor of χ , and $G(y) = 0$. If $f | N$, then $\bar{\chi}(t)$ depends only on $t \bmod N$; the fibres of $(\mathbb{Z}/m)^\times \rightarrow (\mathbb{Z}/N)^\times$ have $\varphi(m)/\varphi(N)$ elements, and substituting $s = tu_y$ gives

$$G(y) = \frac{m}{N} \frac{\varphi(m)}{\varphi(N)} \chi(u_y) \sum_{s \in (\mathbb{Z}/N)^\times} \bar{\chi}(s) (2\langle s \rangle_N - N).$$

For every multiple F of f one has $\sum_{a=1}^F \bar{\chi}(a) a = \frac{F}{f} \sum_{a=1}^f \bar{\chi}(a) a = F B_{1,\bar{\chi}}$ (write $a = a_0 + fk$ with $1 \leq a_0 \leq f$ and use $\sum_{a_0} \bar{\chi}(a_0) = 0$), and $\sum_{s \in (\mathbb{Z}/N)^\times} \bar{\chi}(s) = 0$, as $\bar{\chi}$ is a nontrivial character of $(\mathbb{Z}/N)^\times$. Let N' be the product of the primes dividing N but not f . As $\bar{\chi}(s) = 0$ if $\gcd(s, f) > 1$, Möbius inversion over the divisors e of N' gives

$$\sum_{s \in (\mathbb{Z}/N)^\times} \bar{\chi}(s) (2\langle s \rangle_N - N) = 2 \sum_{e|N'} \mu(e) \bar{\chi}(e) e \sum_{s'=1}^{N/e} \bar{\chi}(s') s' = 2N B_{1,\bar{\chi}} \prod_{p|N'} (1 - \bar{\chi}(p)).$$

Hence $0 = \sum_y c_y G(y) = 2m B_{1,\bar{\chi}}$ times the left side of (8.1). Since $\bar{\chi}$ is odd, $B_{1,\bar{\chi}} \neq 0$, and (8.1) follows. $\square$

Now let $m = 420$. Let χ_4 and χ_3 be the nontrivial characters modulo 4 and 3, ϵ_5 and ϵ_7 the quadratic characters modulo 5 and 7, λ the character modulo 5 with $\lambda(2) = i$, and θ the character modulo 7 with $\theta(3) = \zeta_3 = e^{2\pi i/3}$. The characters χ_4 , χ_3 , ϵ_7 and λ are odd, and ϵ_5 and θ are even. Their values at the primes 2, 3, 5, 7 not dividing the conductor are

	2	3	5	7
χ_4		-1	1	-1
χ_3	-1		-1	1
ϵ_5	-1	-1		-1
ϵ_7	1	-1	-1	
λ	i	$-i$		i
θ	ζ_3^2	ζ_3	ζ_3^2	

For each of the following eleven odd primitive characters χ we apply Lemma 8.4 and divide by a common factor κ of the coefficients. For $x \in B_{420}$, with $s_N = s_N(\chi; x)$, this gives the following relations, each preceded by its character χ :

$$\begin{aligned}
(8.2) \quad & \chi_3: \quad 2s_3 + 4s_6 + 2s_{12} + s_{15} + 2s_{30} + s_{60} = 0; \\
(8.3) \quad & \chi_3\epsilon_5: \quad 3s_{15} + s_{105} = 0; \\
(8.4) \quad & \epsilon_7: \quad 2s_7 + 2s_{21} + s_{35} + s_{105} = 0; \\
(8.5) \quad & \epsilon_5\epsilon_7: \quad s_{35} + 2s_{70} + s_{140} = 0; \\
(8.6) \quad & \chi_4: \quad 3s_4 + 3s_{12} + s_{28} + s_{84} = 0; \\
(8.7) \quad & \chi_4\epsilon_5: \quad s_{20} = 0; \\
(8.8) \quad & \chi_4\chi_3\epsilon_7: \quad s_{84} = 0; \\
(8.9) \quad & \chi_4\chi_3\epsilon_5\epsilon_7: \quad s_{420} = 0; \\
(8.10) \quad & \lambda: \quad 12s_5 + 12(1+i)s_{10} + 6(1-i)s_{15} + 6(1+i)s_{20} + 12s_{30} + 2(1+i)s_{35} \\
& \quad + 6s_{60} + 4is_{70} + 2s_{105} + 2is_{140} + 2(1+i)s_{210} + (1+i)s_{420} = 0; \\
(8.11) \quad & \chi_4\epsilon_5\theta: \quad 2s_{140} + (1 - \bar{\zeta}_3)s_{420} = 0; \\
(8.12) \quad & \chi_4\chi_3\lambda\theta: \quad s_{420} = 0.
\end{aligned}$$

The conductors of these characters are 3, 15, 7, 35, 4, 20, 84, 420, 5, 140, 420, and the factors κ are 24, 4, 8, 4, 16, 12, 4, 1, 2, 1, 1, respectively. For instance, for $\chi = \chi_3$ the factor $1 - \chi_3(7)$ vanishes, so only $N \in \{3, 6, 12, 15, 30, 60\}$ contribute, with $\varphi(420)/\varphi(N) = 48, 48, 24, 12, 12, 6$ and Euler factors 1, 2, 2, 2, 4, 4 (as $1 - \chi_3(2) = 1 - \chi_3(5) = 2$); dividing by 24 gives (8.2). For $\chi = \lambda$ the Euler factors are products of $1 - \bar{\lambda}(2) = 1 + i$, $1 - \bar{\lambda}(3) = 1 - i$ and $1 - \bar{\lambda}(7) = 1 + i$; for $\chi = \chi_4\epsilon_5\theta$ only $N = 140$ and $N = 420$ contribute, with $\varphi(420)/\varphi(140) = 2$ and the factor $1 - \bar{\chi}(3) = 1 - \bar{\zeta}_3$ for $N = 420$, since $\chi(3) = \chi_4(3)\epsilon_5(3)\theta(3) = \zeta_3$. The other cases are the same computation. The ancillary file `level420.py` checks all coefficients, and also checks each of (8.2)–(8.12) on a set of generators of B_{420} .

8.3. Four lemmas.

Lemma 8.5. *For $x \in B_{420}$ the integers*

$$n_{15}(x), \quad n_{35}(x), \quad n_{60}(x), \quad n_{105}(x), \quad n_{140}(x), \quad n_{\mathcal{R}}(x)$$

have the same parity, and $n_{20}(x)$, $n_{84}(x)$, $n_4(x) + n_{12}(x) + n_{28}(x)$ and $n_{420}(x)$ are even.

Proof. For a character χ with values ± 1 one has $s_N(\chi; x) \equiv n_N(x) \pmod{2}$. Reducing (8.2)–(8.9) modulo 2 gives $n_{15} \equiv n_{60}$, $n_{15} \equiv n_{105}$, $n_{35} \equiv n_{105}$, $n_{35} \equiv n_{140}$, $n_4 + n_{12} + n_{28} + n_{84} \equiv 0$, $n_{20} \equiv 0$, $n_{84} \equiv 0$ and $n_{420} \equiv 0$. The 23 divisors $N > 1$ of 420 are the elements of $\mathcal{R}$, the five numbers 15, 35, 60, 105, 140, and 4, 12, 28, 20, 84, 420. Hence $\text{len}(x) = n_{\mathcal{R}} + (n_{15} + n_{35} + n_{60} + n_{105} + n_{140}) + (n_4 + n_{12} + n_{28}) + n_{20} + n_{84} + n_{420}$, and since $\text{len}(x)$ is even, $n_{\mathcal{R}} \equiv 5n_{15} \equiv n_{15} \pmod{2}$. $\square$

Lemma 8.6. *For $x \in B_{420}$ one has $n'_{420}(x) \equiv n_{15}(x) \pmod{2}$.*

Proof. Let $s_N = s_N(\lambda; x) \in \mathbb{Z}[i]$ and $\pi = 1 + i$, so that $2 = -i\pi^2$ and $1 - i = -i\pi$. In (8.10) the coefficients of s_{60} , s_{105} , s_{140} are 6, 2, $2i$, the coefficient of s_{420} is π , and all other coefficients are divisible by 2π . Hence $\pi s_{420} + 2(3s_{60} + s_{105} + is_{140}) \in 2\pi\mathbb{Z}[i]$, and dividing by π ,

$$s_{420} \equiv -(1 - i)(3s_{60} + s_{105} + is_{140}) \pmod{2\mathbb{Z}[i]}.$$

The values of λ are congruent to 1 modulo π , so $s_N \equiv n_N(x) \pmod{\pi}$; as $(1 - i)\pi = 2$, the right side is congruent to $(1 - i)(n_{60} + n_{105} + n_{140})$ modulo 2. On the other hand, $\lambda(y) = \pm 1$

for $y \equiv \pm 1 \pmod{5}$ and $\lambda(y) = \pm i$ for $y \equiv \pm 2 \pmod{5}$. Writing $n' = n'_{420}(x)$ we get $s_{420} \equiv (n_{420} - n') + in' = n_{420} - (1 - i)n' \equiv (1 - i)n' \pmod{2}$, as n_{420} is even by Lemma 8.5. Therefore $(1 - i)(n' - n_{60} - n_{105} - n_{140}) \in 2\mathbb{Z}[i]$, i.e. the integer $n' - n_{60} - n_{105} - n_{140}$ is divisible by π , hence even. By Lemma 8.5, $n_{60} + n_{105} + n_{140} \equiv 3n_{15} \equiv n_{15}$. $\square$

Lemma 8.7. *Let $x = \sum_y c_y(y) \in B_{420}$. For $A \in \{\{\pm 1\}, \{\pm 2\}\}$ and $C \in \{\{\pm 1\}, \{\pm 2\}, \{\pm 3\}\}$ let $n_{A,C}(x)$ be the sum of the c_y over the units $y \in (\mathbb{Z}/420)^\times$ with $y \bmod 5 \in A$ and $y \bmod 7 \in C$. Then for each A the three integers $n_{A,C}(x)$ have the same parity. In particular, if all $c_y \geq 0$ and $n'_{420}(x)$ is odd, then $n_{420}(x) \geq 6$.*

Proof. Let $\chi = \chi_4 \chi_3 \lambda \theta$ and $s_{420} = s_{420}(\chi; x) \in \mathbb{Z}[\zeta_{12}]$, which vanishes by (8.12). We have $\mathbb{Z}[\zeta_{12}] = \mathbb{Z}[i, \zeta_3]$ (as $\zeta_{12} = -i\zeta_3$), and $1, \zeta_3, i, i\zeta_3$ is a $\mathbb{Z}$ -basis of $\mathbb{Z}[\zeta_{12}]$, because $\zeta_3^2 = -1 - \zeta_3$ and $[\mathbb{Q}(\zeta_{12}) : \mathbb{Q}] = 4$. For a unit y one has $\chi(y) = \pm \lambda(y)\theta(y)$, where $\lambda(y) \in \{\pm 1\}$ if $y \bmod 5 \in \{\pm 1\}$ and $\lambda(y) \in \{\pm i\}$ if $y \bmod 5 \in \{\pm 2\}$, and $\theta(y) = 1, \zeta_3^2, \zeta_3$ if $y \bmod 7 \in \{\pm 1\}, \{\pm 2\}, \{\pm 3\}$ (note $\theta(-1) = 1, \theta(2) = \theta(3)^2$). Modulo $2\mathbb{Z}[\zeta_{12}]$ the signs can be ignored, and with $n_{A,C} = n_{A,C}(x)$,

$$\begin{aligned} 0 = s_{420} &\equiv \sum_r r(n_{A_r, \{\pm 1\}} + n_{A_r, \{\pm 2\}}\zeta_3^2 + n_{A_r, \{\pm 3\}}\zeta_3) \\ &= \sum_r r((n_{A_r, \{\pm 1\}} - n_{A_r, \{\pm 2\}}) + (n_{A_r, \{\pm 3\}} - n_{A_r, \{\pm 2\}})\zeta_3) \pmod{2\mathbb{Z}[\zeta_{12}]}, \end{aligned}$$

where r runs over $\{1, i\}$, $A_1 = \{\pm 1\}$ and $A_i = \{\pm 2\}$. Comparing coefficients in the basis $1, \zeta_3, i, i\zeta_3$ gives the first assertion. Now let $c_y \geq 0$ and $n'_{420}(x) = \sum_C n_{\{\pm 2\}, C}$ odd. Then the three $n_{\{\pm 2\}, C}$ are odd, hence positive. As $n_{420}(x)$ is even (Lemma 8.5), $\sum_C n_{\{\pm 1\}, C} = n_{420}(x) - n'_{420}(x)$ is odd as well, so the three $n_{\{\pm 1\}, C}$ are positive, and $n_{420}(x) = \sum_{A,C} n_{A,C} \geq 6$. $\square$

Lemma 8.8. *Let $x = \sum_y c_y(y) \in B_{420}$ with all $c_y \geq 0$.*

- (a) *If $n_{15}(x)$ is odd, then $n_{105}(x) \geq 3$.*
- (b) *$n_{140}(x) \neq 1$.*

Proof. (a) Let $\chi = \chi_3 \epsilon_5$, a character with values ± 1 . Then $s_{15}(\chi; x) \equiv n_{15}(x) \pmod{2}$ is odd, hence nonzero, and (8.3) gives $|s_{105}(\chi; x)| = 3|s_{15}(\chi; x)| \geq 3$. Since $|s_{105}(\chi; x)| \leq n_{105}(x)$, the assertion follows.

(b) Let $\chi = \chi_4 \epsilon_5 \theta$. Its values are sixth roots of unity, so $s_{140} = s_{140}(\chi; x)$ and $s_{420} = s_{420}(\chi; x)$ lie in $\mathbb{Z}[\zeta_3]$. If $n_{140}(x) = 1$, then s_{140} is a root of unity, and (8.11) gives $4 = |2s_{140}|^2 = |1 - \bar{\zeta}_3|^2 |s_{420}|^2 = 3|s_{420}|^2$. This is impossible, since $|s_{420}|^2$ is the norm of s_{420} from $\mathbb{Q}(\zeta_3)$ to $\mathbb{Q}$, an integer. $\square$

8.4. Proof of Theorem 8.1. (i) is Proposition 8.3(c).

(ii) Let $x = \mathbf{u}(a)$. By (i), $n_{15}(x)$ is odd. By Lemma 8.5, $n_{35}(x), n_{60}(x), n_{105}(x), n_{140}(x)$ and $n_{\mathcal{R}}(x)$ are odd, hence at least 1; by Lemma 8.8, $n_{105}(x) \geq 3$ and $n_{140}(x) \geq 3$. By Lemma 8.6, $n'_{420}(x)$ is odd, so $n_{420}(x) \geq 6$ by Lemma 8.7. The sets of elements of order 420, 15, 35, 60, 105, 140 and of order in $\mathcal{R}$ are pairwise disjoint, so a has at least $6 + 1 + 1 + 1 + 3 + 3 + 1 = 16$ entries.

(iii) The first assertion is Proposition 8.3(b),(c). The counts of entries by order, the fact that tw_1 is not a permutation of w_2 for any $t \in (\mathbb{Z}/420)^\times$, and $||w_1|| = ||w_2|| = 96$ are finite computations (level420.py); $\dim M_{w_j} = ||w_j||$ by Section 2.1.

(iv) Let $n \leq 12$ be even and $a \in \mathfrak{B}_{420}^n$. Then a has $n + 2 \leq 14$ entries, so $\mathbf{u}(a) \in L''_{420}$ by (ii), and a is algebraic, as remarked at the beginning of this section. The Hodge conjecture for X_{420}^n follows as in the proof of Corollary 5.4(ii), for odd n because $H^{2k}(X_{420}^n, \mathbb{Q}) = \mathbb{Q}h^k$ for all k . The assertion on products follows from [24, Thm. IV] as in the proof of Corollary 6.1, since only X_{420}^r with even $r \leq 12$ is needed. $\square$

Remark 8.9. (a) By Proposition 8.3, $B_{420} = L''_{420} + \mathbb{Z}\mathbf{u}(w_1)$. Hence, if $w_1 \in \mathfrak{C}_{420}^{14}$, i.e. if the 96-dimensional space $M_{w_1} \subset H^{14}(X_{420}^{14}, \mathbb{Q})$ of Hodge classes is algebraic, then $\mathfrak{C}_{420}^n = \mathfrak{B}_{420}^n$ for every even n by Corollary 5.4(ii). Moreover, in that case the Hodge conjecture holds for X_m^n for every n and every $m = 2^a 3^b 5^c 7^d$: if $420 \mid m$, then $(m/420)_* \xi_{420} \in (m/420)_* B_{420} \subset S_m + \sum_{q \neq 420} \mathbb{Z}\mathbf{u}(\rho_q^{(m)}) + \mathbb{Z}\mathbf{u}((m/420)w_1)$ by Lemma 5.2, and the argument of the proof of Theorem 1.1(C) applies with the additional algebraic characters $t(m/420)w_1$. The same holds with w_2 in place of w_1 .

(b) By Theorem 8.1(iii), for every even $n \geq 14$ the set $\mathfrak{B}_{420}^n$ contains characters a with $\mathbf{u}(a) \notin L''_{420}$ (for instance $w_1 * d$ with $d \in \mathfrak{D}_{420}^{n-14}$), and we do not know whether they are algebraic; so the Hodge conjecture for X_{420}^n with even $n \geq 14$ remains open. The relations (8.2)–(8.12) are specific to the level 420, and we do not treat the levels $m = 420k$ with $k \geq 2$, except for $n = 2$ (Theorem 2.6) and odd n .

APPENDIX A. THE CERTIFICATE

A.1. Statement. Let

$$\begin{aligned} b^* &= 2^{-55} (-18482301002530012 + 5829919155974245 i) \\ &= -0.51298690302653582318015423879842273890972137451171875 \\ &\quad + 0.1618127619666466510128799427548074163496494293212890625 i. \end{aligned}$$

Proposition A.1. *There is a point $y^* \in \mathcal{P}$ with $b(y^*) = b^*$ whose other eleven coordinates differ from the entries of Table 1 by less than 10^{-26} in real and imaginary part, such that:*

- (a) $J_b(y^*)$ is invertible, and y^* is the only point of $\mathcal{P} \cap \{b = b^*\}$ in a box of half-width 2^{-100} containing it;
- (b) y^* is nondegenerate; more precisely $|c_1| \geq 1.974$, $|c_2| = |c_3| \geq 0.156$, $|c_4| \geq 2.073$, $|\text{disc } R| \geq 3.766$, $|\text{Res}(R, P)| \geq 2.567$, $|\text{Res}(R, Q)| \geq 0.419$ at y^* ;
- (c) $\mathcal{I}(y^*, v(y^*)) = 29.588332077128010399 \dots - 6.7688877382028338180 \dots i$, with an error less than 10^{-18} ; in particular $|\mathcal{I}(y^*, v(y^*))| \geq 30.35$.

This implies Proposition 3.17.

coordinate	real part	imaginary part
p_0	0.04254698652900370590236925	0.41959062696659874053932147
p_1	-0.29175162900581561749274186	-0.21152165640381864501207743
p_2	-0.07862929518454702834700365	0.44798979721908385840930498
p_3	-0.32216723198628257586009122	-0.57053846019114427619751128
p_4	0.58751503396883460122333352	0.43478692998204410314107942
q_0	0.41610429362736879506813207	-0.25741447528921419304203880
q_1	-0.49439181483471544169706158	0.03672521494377954466599062
c_1	-0.15591201965849224494420692	-1.96808874336362480139873017
c_2	0.15498354184616354317073508	-0.01894751552276802201975999
c_3	-0.15498354184616354317073508	0.01894751552276802201975999
c_4	-0.42486745291909851785078755	2.02957371846938404761821176

TABLE 1. The coordinates of y^* other than $b = b^*$, to 26 decimals. (On $\mathcal{P}$ one always has $c_3 = -c_2$, the coefficient of s^{10} in (3.2).)

A.2. Krawczyk's test. For a complex box $\mathbf{X} = \prod_{k=1}^N \{z \in \mathbb{C} : |\operatorname{Re}(z - \hat{z}_k)| \leq \varrho, |\operatorname{Im}(z - \hat{z}_k)| \leq \varrho\}$ with centre $\hat{z} = (\hat{z}_k)$, we use the following form of the test of Krawczyk and Moore [18, 20]; see also [21].

Lemma A.2. *Let $\mathcal{F}: \mathbb{C}^N \rightarrow \mathbb{C}^N$ be a polynomial map with Jacobian $D\mathcal{F}$, let $\mathcal{K}$ be an $N \times N$ complex array, and let $\mathbf{J}$ be a set of $N \times N$ arrays, each entry ranging over a complex rectangle, such that $D\mathcal{F}(z) \in \mathbf{J}$ for all $z \in \mathbf{X}$. Suppose that*

$$\hat{z} - \mathcal{K}\mathcal{F}(\hat{z}) + (I - \mathcal{K}J)(z - \hat{z}) \in \operatorname{int} \mathbf{X} \quad \text{for all } J \in \mathbf{J}, z \in \mathbf{X},$$

and that $\|I - \mathcal{K}J\|_\infty < 1$ for all $J \in \mathbf{J}$ (maximum absolute row sum). Then $\mathcal{F}$ has exactly one zero z^ in $\mathbf{X}$, and $D\mathcal{F}(z^*)$ is invertible.*

Proof. For $z \in \mathbf{X}$, $\mathcal{F}(z) - \mathcal{F}(\hat{z}) = \bar{J}(z - \hat{z})$ with $\bar{J} = \int_0^1 D\mathcal{F}(\hat{z} + \theta(z - \hat{z})) d\theta$. Each entry of $\bar{J}$ lies in the closed convex hull of the values of the corresponding entry of $D\mathcal{F}$ on the segment $[\hat{z}, z] \subset \mathbf{X}$, hence in the corresponding rectangle, so $\bar{J} \in \mathbf{J}$. The map $g(z) = z - \mathcal{K}\mathcal{F}(z) = \hat{z} - \mathcal{K}\mathcal{F}(\hat{z}) + (I - \mathcal{K}\bar{J})(z - \hat{z})$ therefore maps the convex compact set $\mathbf{X}$ into itself and has a fixed point z^* by Brouwer's theorem. Since $\|I - \mathcal{K}J\|_\infty < 1$, every $\mathcal{K}J$ with $J \in \mathbf{J}$ is invertible; hence $\mathcal{K}$ is invertible, $\mathcal{F}(z^*) = 0$, and $D\mathcal{F}(z^*) \in \mathbf{J}$ is invertible. If $z, z' \in \mathbf{X}$ are zeros, then $0 = \bar{J}'(z - z')$ with $\bar{J}' = \int_0^1 D\mathcal{F}(z' + \theta(z - z')) d\theta \in \mathbf{J}$ invertible, so $z = z'$. $\square$

A.3. The computation. All computations use the ball arithmetic of Arb [15] (through python-flint) at a working precision of 400 bits; every arithmetic operation returns a ball containing the exact result for all inputs in the input balls. The eleven coefficient polynomials $E_0, \dots, E_{10}$ of (3.2) and all their partial derivatives are computed exactly (with integer coefficients) by computer algebra, checked against an independent forward-mode differentiation of $E_y(s)$, and then evaluated in ball arithmetic.

(K) Fix $b = b^*$ and let $\mathcal{F} = (E_0, \dots, E_{10})$ as a function of the other eleven coordinates. Starting from the approximate coordinates of y^* (Table 1, or the 38-decimal values in the ancillary file), Newton steps (not part of the certificate) produce a centre $\hat{z}$. With $\varrho = 2^{-100}$, $\mathcal{K}$ the midpoint of a numerical inverse of $D\mathcal{F}(\hat{z})$ and $\mathbf{J}$ the ball evaluation of $D\mathcal{F}$ on $\mathbf{X}$, the hypotheses of Lemma A.2 are verified: the image box lies in the interior of $\mathbf{X}$ and $\|I - \mathcal{K}J\|_\infty \leq 4.2 \cdot 10^{-27}$ on $\mathbf{J}$. This gives the point y^* and (a); its coordinates are enclosed in balls of radius less than 10^{-29} , which lie within 10^{-26} of Table 1.

(N) The seven quantities of Definition 3.2 are evaluated at the enclosure of y^* (resultants as Sylvester determinants); all these balls exclude 0, with the lower bounds of (b).

(T) The tangent vector $v^* = v(y^*)$ is the solution of $J_b(y^*)w = -\partial\mathcal{F}/\partial b$, $v_b^* = 1$. Solving this linear system in ball arithmetic with coefficients evaluated on the enclosure of y^* gives an enclosure of v^* with radius below 10^{-48} .

(I) The residue sum $\mathcal{I}(y^*, v^*)$ of Definition 3.4 is evaluated in ball arithmetic: the roots $\rho_{1,2} = (-b^* \pm (b^{*2} - 4)^{1/2})/2$ are enclosed, and $\operatorname{Res}_{s=\rho_1} \eta_v$ is the coefficient of h in the Taylor expansion of $(\rho_1 + h - \rho_2)^{-2} P\Delta_v / (c_3 c_4 s)$ at $s = \rho_1 + h$, computed with truncated power series; similarly for ρ_2 . The result is the ball of (c). As an independent check, which is not part of the certificate, the script also evaluates $\mathcal{I}(y^*, v^*)$ in floating point by numerical integration of η_v over small circles around ρ_1 and ρ_2 ; the two values agree to within 10^{-39} .

A.4. Ancillary files. The ancillary files of this paper contain `certify.py`, which performs the computation of Section A.3 and compares its results with Table 1 and Proposition A.1; `data/point.json`, which contains b^* and the coordinates of y^* to 38 decimals; `identities.py`, which checks the polynomial identities of Section 3, the identities of Appendix B and the other finite computations with characters used in Sections 2–4 and 7; `gap.py`, which carries out the computations of Section 5 (Lemma 5.7, Proposition 5.8 and Remarks 5.10 and 7.5); `level420.py`,

which carries out the computations of Section 8; a `README.md` describing the checks; and the expected output of the four scripts in `expected_output/`. `certify.py` requires Python 3 with `sympy`, `mpmath` and `python-flint`, and `identities.py` requires Python 3 with `sympy`; `gap.py` and `level420.py` use only the Python standard library, and `python-flint` only for the independent check of the indices in the proofs of Propositions 5.8 and 8.3. Each script runs in about a minute or less.

APPENDIX B. THE IDENTITIES

B.1. The characters α and κ . Table 2 lists $t\alpha$ and $t\kappa$ (entries in $\{1, \dots, 34\}$) with $|t\alpha|$ and $|t\kappa|$; it proves Lemma 3.1(a),(b) and Lemma 4.1. Lemma 3.1(c) is checked directly: the ten sums of two entries of α are 3, 17, 22, 31, 18, 23, 32, 2, 11, 16 mod 35.

t	$t\alpha$	$t\kappa$	$ t\alpha $	$ t\kappa $	t	$t\alpha$	$t\kappa$	$ t\alpha $	$ t\kappa $
1	(1, 2, 16, 21, 30 17, 22, 31)		2	2	8	(8, 16, 23, 28, 30 31, 1, 3)		3	1
2	(2, 4, 32, 7, 25 34, 9, 27)		2	2	13	(13, 26, 33, 28, 5 11, 6, 18)		3	1
3	(3, 6, 13, 28, 20 16, 31, 23)		2	2	16	(16, 32, 11, 21, 25 27, 2, 6)		3	1
4	(4, 8, 29, 14, 15 33, 18, 19)		2	2	17	(17, 34, 27, 7, 20 9, 24, 2)		3	1
6	(6, 12, 26, 21, 5 32, 27, 11)		2	2	23	(23, 11, 18, 28, 25 6, 16, 13)		3	1
9	(9, 18, 4, 14, 25 13, 23, 34)		2	2	24	(24, 13, 34, 14, 20 23, 3, 9)		3	1
11	(11, 22, 1, 21, 15 12, 32, 26)		2	2	26	(26, 17, 31, 21, 10 22, 12, 1)		3	1
12	(12, 24, 17, 7, 10 29, 19, 22)		2	2	29	(29, 23, 9, 14, 30 3, 8, 24)		3	1
18	(18, 1, 8, 28, 15 26, 11, 33)		2	2	31	(31, 27, 6, 21, 20 2, 17, 16)		3	1
19	(19, 3, 24, 14, 10 8, 33, 29)		2	2	32	(32, 29, 22, 7, 15 19, 4, 12)		3	1
22	(22, 9, 2, 7, 30 24, 29, 17)		2	2	33	(33, 31, 3, 28, 10 1, 26, 8)		3	1
27	(27, 19, 12, 7, 5 4, 34, 32)		2	2	34	(34, 33, 19, 14, 5 18, 13, 4)		3	1

TABLE 2. The conjugates of α and κ . The left half is $t \in \mathcal{T}$.

B.2. The identity for γ . Both sides of Proposition 7.1 are the multiset of 22 elements of $\mathbb{Z}/70$

$$(B.1) \quad \{1, 2, 4, 10, 17, 18, 20, 24, 26, 32, 34, 36, 38, 42, 44, 52, 53, 60, 61, 62, 66, 68\}.$$

Indeed

$$\begin{aligned} \gamma \cup \varepsilon &= \{1, 20, 24, 42, 61, 62\} \cup \{2, 4, 10, 17, 18, 26, 32, 34, 36, 38, 44, 52, 53, 60, 66, 68\}, \\ 2\beta \cup \varsigma_1 \cup \varsigma_2 \cup \sigma_3 &= \{2, 4, 32, 34, 42, 44, 60, 62\} \cup \{1, 18, 53, 68\} \\ &\quad \cup \{17, 26, 36, 61\} \cup \{10, 20, 24, 38, 52, 66\}. \end{aligned}$$

The conditions $|t\varsigma_1| = |t\varsigma_2| = 2$ and $|t\gamma| = 3$ for all $t \in (\mathbb{Z}/70)^\times$, and $|t\sigma_{5,5}| = 3$ for all $t \in (\mathbb{Z}/35)^\times$ (which also follows from [2, p. 387]), are verified by `identities.py`. The memberships $2\beta \in \mathfrak{B}_{70}^6$ and $\sigma_3 \in \mathfrak{B}_{70}^4$, needed for Theorem 2.4(i), follow from $\beta \in \mathfrak{B}_{35}^6$ and $\sigma_{5,5} \in \mathfrak{B}_{35}^4$ by Lemma 2.3(c), since $|t(ea)| = |\bar{t}a|$; the first is also checked directly by `identities.py`. The identities $\varsigma_1 * (36, 34, 35, 35) \sim \sigma_{2,18} * \sigma_{2,1}$ and $\varsigma_2 * (52, 18, 35, 35) \sim \sigma_{2,26} * \sigma_{2,17}$ at level 70, and $\sigma_3 \sim \sigma_{5,10}$, are checked there as well.

B.3. The identity for δ . Both sides of Proposition 7.3 are the multiset

$$\{3, 24, 60, 72, 94, 126, 138, 164, 183, 186\} \subset \mathbb{Z}/210;$$

one checks $47 \cdot (2, 9, 129, 142, 168, 180) \equiv (94, 3, 183, 164, 126, 60) \pmod{210}$, and $|t\delta| = 3$, $|t\tau| = 2$ for all $t \in (\mathbb{Z}/210)^\times$; moreover $\tau = (24, 24 + 70, 24 + 140, 210 - 3 \cdot 24)$ is the standard character $\sigma_{3,24}$ up to order.

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